

ARGUS Morphing Wing: High-Fidelity RANS Validation of the VLM Design Study

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Abstract

Six wing surfaces were meshed and solved with steady RANS. One is the rigid baseline. Five are morphing candidates a VSPAERO vortex-lattice study selected on induced drag alone. Every geometry is trimmed to its own condition's target lift. All coefficients are on the DSO basis, $S_{\text{ref}} = 1.24092 \text{ m}^2$ and $C_{\text{ref}} = 0.39396 \text{ m}$. Ninety cases converged and all eighteen trims are solved. At Condition CR, $M = 0.10$ and $C_L = 0.428277635108$, the candidates span -0.868 to $+0.232$ counts against a baseline of 188.031 counts, four of the five below it. At early cruise, $M = 0.78$ and $C_L = 0.529297087450$, W saves 0.769 counts against a baseline of 177.839, F and H cost 1.796 and 2.244, and C and M cost 12.871 and 14.554. Late cruise, $M = 0.78$ and $C_L = 0.544114800210$, gives the same order, W at -0.781 to M at $+13.728$ against 187.959, and W is the cheapest geometry at all three operating points. The cost follows the aft shock: C and M carry a strong shock at $\eta = 0.80$ and separate behind it, F and H a moderate one and stay attached, W none. The far-field induced-drag metric the study now ranks on reproduces the measured Condition CR ordering to nine pairs of ten; the near-field metric the June selection was made on reproduces four of ten, and neither resolves the size of the effect. Every candidate raises the hinge moment about the morph line, by 0.4 to 6.3 per cent, and the root-bending constraint that selected C is not satisfied once computed viscously: C moves from $+6.796$ per cent in the vortex-lattice study, inside the 6.8 per cent screen, to $+7.170$ per cent in RANS, outside it. The hinge moments tabulated for the hinged and twist concepts are common-basis comparisons rather than the load their own mechanisms carry, so the two morphing concepts cannot yet be ranked on actuator load.

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Nomenclature

Reference quantities are dual and are never mixed. The DSO basis is primary: all three fixed-lift targets are defined on it and every RANS coefficient here is on it, measured from each case's own `forceCoeffs` header rather than declared. The TP-1580 tunnel basis is reported alongside. The two areas differ by $13.3572/12.0 = 1.113100$, i.e. 11.31 per cent, against a morphing effect of order 0.4 per cent, so a missed conversion is about 28 times the signal. The exact reciprocal 0.898392 carries a TP-1580 coefficient onto the DSO basis, and these two are the pair used throughout; 1.113132 is the same ratio formed from the rounded metric areas and is not the reciprocal of 0.898392.

Symbol	Definition, value where fixed, and frame
$S_{\text{ref}}, C_{\text{ref}}, c_{\text{ref}}$	Full-wing reference area and chord: DSO 1.24092 m ² (13.3572 ft ²) with $C_{\text{ref}} = 0.39396$ m; TP-1580 1.1148 m ² (12.0 ft ²) with $c_{\text{ref}} = 0.3404$ m, the chord used for Reynolds matching to the tunnel
$A_{\text{ref}}, l_{\text{ref}}$	Half-wing reference area carried by every converged case, 0.620462 m ² ($2A_{\text{ref}}$ reproduces the DSO S_{ref} to 3.22×10^{-6} relative), and the mean aerodynamic chord 0.393957 m
AR, b_{ref}	Aspect ratio 10.7807 on the DSO area and 12.0000 on TP-1580's; reference span 3.6576 m (12.0 ft), common to both
C_L, C_D	Lift and total drag coefficient, each a 200-sample gate window mean, never a last sample, on the DSO half-model basis, patch wing
C_M	Pitching-moment coefficient about (1.34, 0, 0) m, nose-up positive; written by forceCoeffs and reported per case, with the trim set carrying forces only (Table 13)
$L/D, \text{ct}$	C_L/C_D , both window means, the reference area cancelling; drag count, 1×10^{-4} in a force coefficient. A C_L residual in counts is written “counts of C_L ”
ΔC_D	Candidate-minus-baseline drag increment at fixed C_L , within one condition only
$C_{D,p}, C_{D,v}$	Pressure and viscous drag split; closes at Condition CR and, once q carries ρ , at cruise
C_{Di}, C_{Diw}	Induced drag, near field (surface integration) and far field (wake plane), both VSPAERO columns. The DSO reranked onto C_{Diw} on 19 August 2026. The RANS Trefftz-plane induced drag, and the span efficiency it implies, are in Section 5.5
C_{Dtot}, C_{Do}	VSPAERO total $C_{Do} + C_{Di}$, held apart from any RANS total; and its flat-plate skin-friction correlation, whose $Re = 1 \times 10^7$ is a VSPAERO placeholder entering only that correlation, the VLM itself being inviscid
e, E, E_w	Span efficiency in $C_{Di} = C_L^2/(\pi AR e)$; E near field and E_w far field as printed by VSPAERO
c_p, c_p^*	Pressure coefficient $(p - p_\infty)/q$, formed in the solver so the normalisation travels with the file; and the sonic value -0.4931 , derived at $M = 0.780302$ with $\gamma = 1.4$. These carry no reference area
$c_f, c_{f,s}$	Skin friction $ \tau_w /q$, and its signed freestream-aligned form $m(\tau_w \cdot \hat{d})/q$ with $m = -1$ calibrated per case, where $\tau_s < 0$ marks reversed flow
$c_l, c_l c/C_{\text{ref}}$	Sectional lift coefficient (pressure only) and sectional load, the spanwise loading quantity
m'_h, c_{mh}	Sectional hinge moment per unit span, N m/m, and its coefficient $m'_h/(q_\infty c(y)^2)$ formed on the <i>local</i> section chord rather than a reference chord. Taken about the swept $x_h/c = 0.62$ line, projected on that line's <i>local tangent</i> , over aft surfaces only, positive trailing-edge down
M_x	Half-wing root bending moment about the root chord line at $y = 0$, N m, taken from the solver's own moment output rather than integrated from a surface. Dimensional, so it is <i>not</i> comparable across dynamic pressures; the comparison against the low-order method is made on the percentage increase over each method's own baseline, which cancels q (Section 5.7)
$M_h, C_{Mh, \text{region}}$	Integrated half-wing hinge moment over $0.60 \leq \eta \leq 1.00$, N m, and its coefficient $M_h/[q_\infty \int c(y)^2 dy]$ over the same region, which avoids selecting one of this wing's several defensible reference chords. M_h is dimensional and therefore <i>not</i> comparable across the three dynamic pressures; $C_{Mh, \text{region}}$ is
C_f	Flat-plate correlation $0.026 Re^{-1/7}$, used only in the y^+ stack design and distinct from c_f
M, α, α_{0L}	Mach number; incidence in degrees read from the solver's own drag-direction header, never from the case name; zero-lift angle from the bracket fit
$U_\infty, \rho_\infty, p_\infty, T_\infty, q$	Freestream speed, density, static pressure, temperature and dynamic pressure; q is kinematic at Condition CR ($\frac{1}{2} U ^2 = 578 \text{ m}^2/\text{s}^2$) and absolute at cruise
γ	Ratio of specific heats: 1.4 for c_p^* , and the file's own 1.4010834 for the late-cruise dynamic-pressure closure
$\nu, \mu, \nu_t, \tilde{\nu}$	Kinematic and dynamic viscosity (the cruise μ is the model-scale Reynolds-matched value, not the physical one); eddy viscosity $\nu_t = k/\omega$; Spalart-Allmaras transported variable
k, ω, Tu	Turbulence kinetic energy, specific dissipation rate, and freestream turbulence intensity $\sqrt{2k/3}/U_\infty$
$Re_{C_{\text{ref}}}, Re_{c_{\text{ref}}}$	Reynolds number on the DSO and on the TP-1580 chord respectively; the subscript names which chord, and both are quoted for every condition
$\tau_w, \tau_s, u_\tau, y^+$	Wall shear stress, its signed streamwise component, friction velocity, and $y u_\tau/\nu$, quoted throughout on the <i>cell-centre</i> convention the solver evaluates, on which a stack designed at the full first-cell height reads half its nominal target (Section 3.1)
$\eta, x/c, x_h/c$	Spanwise station on $b_{\text{ref}}/2$ unless a table names the extractor axis, which is 0.19 per cent inboard of it; chordwise fraction of local chord; morph start or hinge line, 0.62 continuous and 0.70 hinged
$\xi, S(\xi), A/c, \Delta z_c, \Delta\theta$	Morph coordinate $(x/c - x_h/c)/(1 - x_h/c)$; smoothstep $3\xi^2 - 2\xi^3$; commanded camber amplitude on the local chord, positive trailing-edge down; camber displacement $-A(\eta)S(\xi)$ applied equally to both surfaces so thickness is preserved; commanded twist in degrees
W, n, τ	Convergence-gate window (200 force samples); converged bracket points in a fit; Kendall rank correlation (wall shear is τ_w)
B, C, F, M, W, H	The six geometries: rigid baseline; continuous trailing-edge camber (C, F, M); distributed twist (W); conventional hinged (H)
B*	The corrected-interpolation baseline the DSO ranks against, which no RANS case has ever meshed
Condition CR	The $M = 0.10$ low-speed state the design study selected its candidates at. CR does not stand for cruise: the two cruise states are at $M = 0.78$ and are always written “early cruise” and “late cruise”. Unqualified “cruise” in this report means those two and never Condition CR
SW*, CMP*, LC*	Case prefixes at Condition CR, early cruise and late cruise respectively
DSO, VLM, RANS, EET	Design-space optimisation, the VSPAERO vortex-lattice study that selected the candidates; vortex-lattice method; Reynolds-averaged Navier-Stokes; Energy Efficient Transport, the NASA TP-1580 high-aspect-ratio supercritical-wing model

1 Objectives and the question

Purpose. ARGUS asks how the outer wing of a transport aircraft should adapt during cruise. A low-order VSPAERO study searched that design space and selected the candidate shapes [4]. The search is inviscid,

wing-only, and ranks on induced drag. This work is the viscous audit of that search.

Study hierarchy. The design study evaluates hundreds of geometries cheaply. This campaign evaluates six of them with steady RANS at the matched Reynolds number. The design study proposes candidates. The RANS says what they cost at trimmed lift.

The job. Six wing surfaces, one rigid baseline and five morphing candidates, were meshed and solved with steady RANS at three operating points, every geometry trimmed to that point’s target lift. The low-order study selected the candidates at the low-speed point, Condition CR, so that is where it is tested against the RANS; the two cruise points are the results the low-order study does not have. **Question.** This report answers two. First, does the low-order method order the candidates the way the RANS orders them? Second, what does the RANS say about the candidates themselves?

Reading order. The report follows the order of the work. Section 2 states the operating points and the six geometries. Section 3 states the OpenFOAM setup, the automated pipeline, the trim convention and the convergence gate, each once. Section 4 lists every quantity of interest, its frame, and whether the low-order study produces it. Section 5 reports each quantity in turn, RANS beside VLM wherever a VLM value exists. Section 6 answers whether the VLM can drive the design study, and Section 7 states what the whole of it establishes and what it does not. Appendices A and B carry the surface pressure and skin friction, by station and over the whole wing.

2 Conditions and geometries

Purpose. This section fixes the three states every number in this report is read at, and says what was changed on the wing. Five candidates are compared against one rigid baseline. Each moves a different part of the outer wing by a different amount. Every geometric number here is read from the design files and the delivered section schedules, and mesh statistics are quoted where they bear on the geometry.

2.1 Operating points

Purpose. Three states are fixed here, and every number in this report is read at one of them. Two are cruise states of the ARGUS aircraft at $M = 0.78$. One is the low-speed point the DSO optimisation selected its candidates at. Table 2 is the record of what ran. **“Condition CR” is the low-speed point, and CR does not stand for cruise.** It is the $M = 0.10$ state the design study selected its candidates at, and it is named Condition CR throughout this report and in the delivered packages. The two states that *are* cruise are always written “early cruise” and “late cruise”, both at $M = 0.78$. The distinction is load-bearing and the wrong reading inverts results: where Section 5.6 reports a penalty that is smaller at Condition CR than at early cruise, reading CR as cruise makes the sentence contradict itself. Wherever this report says “cruise” without qualification it means the two $M = 0.78$ states and never Condition CR.

Method. Each cruise state is pinned by mass, altitude and Mach number. Standard-atmosphere density and speed of sound follow from the altitude. The target lift coefficient is then $C_{L,\text{target}} = W/(q_\infty S_{\text{ref}})$. Every value in Table 2 is read from the recipes’ `CONDITION.json` and from each built case’s own `forceCoeffs` block.

Scale. The cruise states are defined at aircraft scale, on a chord of 3.196 m. The meshed geometry is model scale, on $C_{\text{ref}} = 0.393957$ m. Viscosity is set so that Reynolds number on the model chord equals the aircraft-scale value. The table therefore carries a model-scale μ beside an aircraft Reynolds number.

Table 2: The three operating points. Coefficients are on the DSO basis, $S_{\text{ref}} = 1.24092 \text{ m}^2$ and $C_{\text{ref}} = 0.39396 \text{ m}$; the RANS cases solve a half wing at $A_{\text{ref}} = 0.620462 \text{ m}^2$. Reynolds number is on the DSO chord: at Condition CR it is 0.79 million on the TP-1580 chord of 0.3404 m, which is the figure in wider circulation. Condition CR is solved incompressible, so its pressure is kinematic and its temperature is not defined; the tabulated q_∞ is $\frac{1}{2}\rho_\infty|U_\infty|^2$ on the $\rho_\infty = 1.225$ its `forceCoeffs` block carries. Our cruise q_∞ runs 0.303 per cent above Zheng’s Table 18 [4] at both states, common mode within our runs.

	Condition CR	Early cruise	Late cruise
Case prefix	SW*	CMP*	LC*
Flight state	DSO-matched	74,500 kg, 10 km	56,000 kg, 12 km
Mach	0.10	0.78	0.78
Target C_L	0.428277635108	0.529297087450	0.544114800210
$ U_\infty $ (m/s)	34.0	233.934028	230.501789
ρ_∞ (kg/m ³)	1.225	0.41271	0.31083
T_∞ (K)	n/a	223.15	216.65
p_∞ (Pa)	n/a (gauge)	26,495.90	19,373.96
q_∞ (Pa)	708.05	11,292.80	8,257.37
Viscosity	$\nu 1.46 \times 10^{-5}$	$\mu 1.7961143 \times 10^{-6}$	$\mu 1.7523444 \times 10^{-6}$
$Re_{C_{\text{ref}}}$	0.92×10^6	2.1176454×10^7	1.6107442×10^7
Solver	incompressible	compressible	compressible
Turbulence model	kOmegaSST	Spalart–Allmaras	Spalart–Allmaras
Wall treatment	resolved	modelled	modelled

Why the two cruise states differ. The aircraft burns fuel and steps from 10 to 12 km, so mass falls by a quarter and density falls with the altitude. Dynamic pressure falls faster than weight, so the required lift

coefficient rises by 2.80 per cent while Reynolds number drops by 24 per cent.

The two states close against each other. At fixed reference area the ratio of targets equals the mass ratio over the dynamic-pressure ratio. The targets give 1.0279951. Mass and dynamic pressure give 1.0279974, agreeing to 2.3×10^{-6} relative. The two points are one flight state read at two masses.

2.2 The wing every candidate starts from

Baseline. All six surfaces are the same NASA EET AR-12 cruise wing. Only the outer wing differs between them. Fuselage, tail and high-lift system are not modelled. Table 3 gives the planform an experimentalist would need.

Section. The aerofoil is a supercritical section of 11.97 per cent thickness. Its trailing edge is blunt at 0.63 per cent of local chord. That edge is 1.03 mm at the tip and sets the surface cell size everywhere.

Table 3: Baseline planform, read from the geometry sidecar of the surface the cases mesh, `geometry/derived/wing3d/baseline.oaml_placed.json`. Measured values are taken from the lofted surface; targets are the registered references. The half-wing planform area integrates to 0.620463 m^2 against the $A_{\text{ref}} = 0.620462 \text{ m}^2$ the cases carry. The meshing surface adds a 20 mm root extension through the symmetry plane as scaffolding, which is cut away and is not part of the wing.

Quantity	Measured	Target	Note
Root chord (m)	0.639125	0.640081	
Tip chord (m)	0.152172	0.152400	taper ratio 0.238
Semispan (m)		$b_{\text{ref}}/2 = 1.8288$	half model on a symmetry plane
Leading-edge sweep (deg)	28.8447	28.8620	quarter chord 27
Tip minus root twist (deg)	-4.129406	-4.11495	washout
Root incidence (deg)	1.852649	1.873	placement transform
Dihedral rise over semispan (m)	0.130498	0.139	placement transform
Half planform area (m^2)		0.620463	integrated from the section table
Thickness, trailing edge	11.97 % chord; TE 0.63 % local chord		TE 1.03 mm minimum

Span coordinate. Stations below are quoted in $\eta = y/(b/2)$. The wing is lofted through 19 defining stations. Those stations carry the morphing, and every realised value in this section is read at them.

2.3 Two lineages of the same wing

Why there are two. The delivered design file lofts the wing through 19 stations but carries only 9 distinct aerofoils: ten stations inherit a neighbour's section instead of a blended one, so thickness is held constant across runs of stations and then steps. A corrected file was issued afterwards in which all 19 stations carry their own aerofoil. The meshed baseline is the uncorrected file. All five candidates are built on the corrected one.

What the difference is. Figure 1 shows it. Both wings are the same planform at the same twist, so the difference is thickness and nothing else. They agree exactly at every station both files define and differ only between those stations, which is why the uncorrected thickness distribution is a staircase and the corrected one a smooth taper. The difference is zero at and inboard of $\eta = 0.450$. The first station that differs, $\eta = 0.515$, is also the largest, at 0.407 percentage points of t/c and 0.209 per cent of chord in surface ordinate. Camber differs by at most 0.0034 per cent of chord, two orders of magnitude below the thickness difference.

What it costs the comparison. Nothing for the candidate ranking. All five candidates share the corrected lineage, so the difference is common to both sides of every candidate-to-candidate number in this report and cancels from it. It enters only where a candidate is differenced against the meshed baseline, and there it is bounded by the figures above.

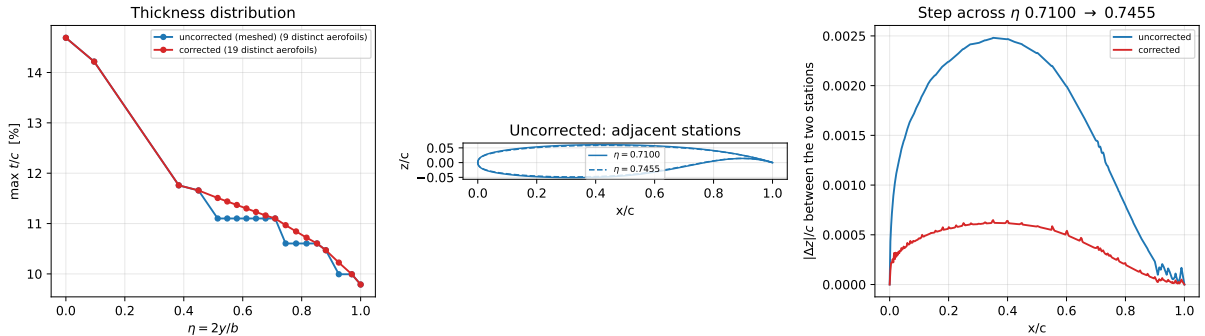


Figure 1: The two lineages of the baseline wing. Left: maximum t/c against η , the uncorrected file (9 distinct aerofoils across 19 stations) against the corrected file (19). The uncorrected curve holds thickness constant across runs of stations and steps between them; the corrected curve tapers. Centre: the two adjacent uncorrected stations either side of the largest step, overlaid, which is what an inherited rather than blended section looks like. Right: the surface ordinate difference $|\Delta z|/c$ across that step for each lineage; the uncorrected step reaches 0.25 per cent of chord where the corrected reaches 0.06. Measured from the two design files by `scripts/measure_lineage_pair.py` into `data/lineage_measured.json`.

2.4 How the morphing is commanded

Three mechanisms appear across the five candidates. Each acts on a different part of the section.

Continuous trailing-edge camber (C, F, M). Camber changes behind a fixed start line at $x_h/c = 0.62$. Let

$$\xi = \frac{x/c - x_h/c}{1 - x_h/c}, \quad 0 \leq \xi \leq 1. \quad (1)$$

The camber displacement follows a smoothstep in ξ ,

$$S(\xi) = 3\xi^2 - 2\xi^3, \quad \Delta z_c(x, \eta) = -A(\eta) S(\xi), \quad x/c \geq x_h/c. \quad (2)$$

Both S and its slope vanish at the start line. The moving surface therefore leaves the fixed forward aerofoil without a step or a slope break. Upper and lower surfaces are displaced by the same increment, so thickness is preserved by construction. The delivered sections carry zero change in t/c and a trailing-edge displacement equal to $-A/c$ to 10^{-9} . Positive A is trailing edge down. The amplitude A/c is normalised on the *local* chord.

Distributed twist (W). The whole local section rotates about $x/c = 0.25$. The aerofoil profile is unchanged and only incidence moves. The command is an incremental twist in degrees, added to the baseline washout. This is the one candidate whose leading edge moves.

Conventional hinged (H). Two aileron-like segments rotate rigidly about a hinge at $x_h/c = 0.70$. Deflection is constant within a segment and steps between them. The slope breaks at the hinge, which is what separates this concept from camber morphing. It is modelled as a zero-gap outer mould line, with no gap, seal or bracket represented.

Command layout. Each continuous concept carries five spanwise commands. These are design variables and not geometric sections. Values between them are interpolated onto the 19 defining stations before the surface is lofted. Table 4 gives the commands as the design files hold them.

Table 4: The six geometries and the command each flies, read from the DSO design files (`<case>_design.json`) and `registry/candidates.yaml`. Camber commands are A/c on the local chord at $\eta = 0.60, 0.70, 0.80, 0.90, 1.00$. Twist commands are $\Delta\theta$ in degrees at $\eta = 0.68, 0.76, 0.84, 0.92, 1.00$, positive leading edge up. The hinged command is the inboard and outboard segment deflection in degrees, positive trailing edge down. The STL column is the SHA-256 prefix of the placed surface the cases mesh. All five candidates were selected at Condition CR, $M = 0.10$, $C_L = 0.428277635108$ (Table 2).

ID	DSO case	Concept	Command	STL
B	<code>baseline.wing.only.refined</code>	rigid	none	20bd6acd
C	<code>cte_i002.c04</code>	TE camber	0.016427, 0.021826, 0.033726, 0.015955, 0.000015	790c3ded
F	<code>cfft_b02.c01</code>	TE camber	0.012607, 0.013869, 0.012069, 0.012284, 0.000461	82bf2e1e
M	<code>mcv2_i002.c01</code>	TE camber	0.008027, 0.025978, 0.034968, 0.018199, 0.000218	dae22d0c
W	<code>cffw_b01.c01</code>	twist	+1.7997, +1.1260, +1.2782, +0.2573, -0.9994	200fa64c
H	<code>chc_g02.c06</code>	hinged	2.4954 inboard, 1.6636 outboard	18329aa3

Two generations, not five designs. C and M come from the July selection. F, W and H come from the August reselection, after the design study moved from near-field to far-field induced drag. C and F are the same concept ranked under two metrics. F’s peak amplitude is 41 per cent of C’s.

2.5 What each candidate actually moves

Method. Commands are interpolated onto the defining stations, so the realised peak sits at a station rather than at a control point. The chord tapers by a factor of 4.2 from root to tip. A constant A/c therefore produces a falling physical displacement outboard. Table 5 gives commanded amplitude and realised displacement together, and Figure 2 shows the same thing on the wing.

Table 5: Realised geometry change at the 19 defining stations. Computed from the delivered schedules: `<case>_section.schedule.csv` for C, F, M and H, and `cffw_b01.c01.twist.schedule.csv` for W. “Peak command” is the largest amplitude reached at a defining station. It is A/c for the camber candidates and degrees for W and H. “Peak TE” is the largest trailing-edge displacement in millimetres, with the station it occurs at. For W the trailing edge travels on an arc, and the entry is $0.75c \sin \Delta\theta$ derived from the tabulated twist and chord. Displacement is trailing edge down in every case except W outboard of $\eta = 0.95$.

ID	Acts over	Stations	Peak command	at η	Peak TE (mm)	at η
C	η 0.6125 to tip	12	0.031465	0.781	6.894	0.781
F	η 0.6125 to tip	12	0.013689	0.710	3.452	0.6125
M	η 0.6125 to tip	12	0.033260	0.781	7.288	0.781
W	η 0.6125 to tip	12	+1.7435 deg	0.6775	5.719	0.6775
H	η 0.710 to 0.970	8	2.4954 deg	0.710	3.145	0.710

Result. Every candidate moves the wing by millimetres. The largest movement anywhere in the set is 7.29 mm, on M. That is 1.1 per cent of the root chord. C follows at 6.89 mm and W at 5.72 mm. F and H are the smallest at 3.45 and 3.15 mm.

Each concept on the wing: where the surface actually moves, seen from above and below

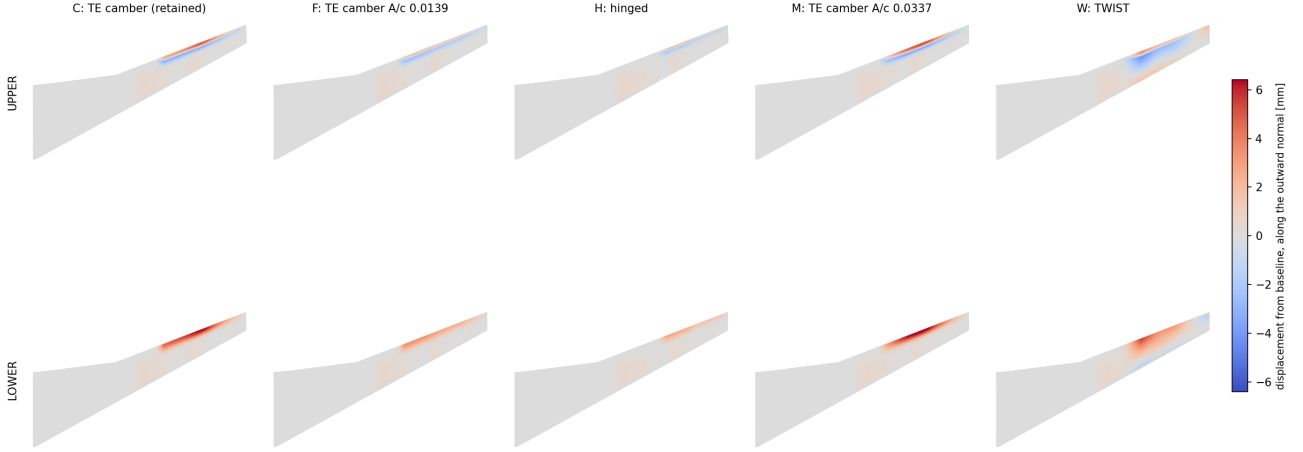


Figure 2: Where each concept moves the surface, seen from above and below. Panel letters are the geometry IDs of Table 4. Span runs left to right with the root at left and chord up the page. Colour is displacement from the baseline surface measured along the baseline’s own outward normal, so the same colour means the same physical direction on both surfaces: the four camber concepts move the lower surface down and the upper surface up over the aft wing, and W, a rigid rotation about $x/c = 0.25$, moves the whole section including the leading edge, which is the signature that distinguishes a twist from a camber morph. One colour scale spans every panel, ± 6.42 mm at the 99.5th percentile, because per-panel autoscaling would make five different magnitudes look alike. Surfaces are matched by nearest point rather than by vertex index, since the three later concepts carry 195,266 triangles against the baseline’s 192,588. Generated by `scripts/render.concept.3d.py`.

Taper moves the peak. F shows the effect plainly. Its command peaks at $\eta = 0.710$, but its physical displacement peaks at $\eta = 0.6125$, the innermost morphed station. The chord there is 0.270 m against 0.241 m. C and M peak in both measures at the same station, because their commands rise steeply enough to beat the taper.

The tip is where the concepts part. All three camber candidates close to near zero at the tip: 0.002 mm on C, 0.033 mm on M and 0.070 mm on F. W does the opposite. It reaches $+1.74$ deg of washin at $\eta = 0.6775$, crosses zero between $\eta = 0.926$ and 0.970 , and ends at -0.999 deg of washout at the tip. W is the only bidirectional candidate in the set, and the only one that unloads the tip by rotating it.

H is discrete where the others are smooth. Its inboard segment holds 2.4954 deg from $\eta = 0.710$ to 0.852. Its outboard segment holds 1.6636 deg to $\eta = 0.970$. Deflection steps by 0.832 deg at the segment break. The outer 3 per cent of span carries no deflection at all.

2.6 Where the morphing acts in span

Region. The continuous concepts act over $\eta = 0.60$ to the tip. That interval is 25.2 per cent of the half-wing planform, taper making it far less than the 40 per cent of span it occupies. Only the chord aft of $x_h/c = 0.62$ moves, so the moving surface is 9.6 per cent of the half-wing planform. H acts over $\eta = 0.710$ to 0.970, which is 15.4 per cent of planform, and moves 4.6 per cent aft of its hinge.

Measured effect. Where that geometry appears in the loading is shown in Section 5.4, Figure 9: the difference from baseline sits over the morphed span in every case.

2.7 Meshed surfaces and their statistics

Route. All six surfaces are lofted by one procedure, step for step, but that procedure was run twice: B, C and M went through it in July and F, W and H in August, and the two runs differ only in the tip-cap construction. Each is tessellated at 200 chordwise points per surface and 240 spanwise stations, with the 19 defining stations retained. The same placement transform and the same blunt-trailing-edge convention follow. Table 6 gives the surfaces and the meshes built on them.

What the mesh size tracks. Nothing but the condition: the six Condition CR meshes lie within 0.10 per cent of one another and the twelve cruise meshes within 0.15 per cent, so every candidate-minus-baseline difference is taken between meshes of the same size.

The integrated surface. The wing patch carries 5,276,562 to 5,277,100 faces across the eighteen trim cases, a spread of about one face in 10^4 . The surface discretisation is therefore common to both sides of every candidate-minus-baseline difference. The meshing surface also carries the 20 mm root extension, so its STL area of 1.3497 m² stands 5.49 per cent above the wing’s wetted area of 1.2794 m². Wetted area is taken from the solver wing patch.

Table 6: Meshed surfaces and as-built mesh size, on the meshes every result in this report is taken from. Triangle counts are from each surface’s own sidecar and split by delivery, not by concept. Condition CR counts are the `checkMesh` figures carried by the run cards; cruise counts are from each mesh’s own `polyMesh/owner` header. The surface gate is `surfaceCheck` on the case copy: zero open edges, zero over-connected edges, one zone and one normal orientation. Mesh recipe and layer stacks are in Section 3.1. Source: rows generated by `scripts/make_cruise_tables.py`.

ID	Concept	Triangles	Cells, Condition CR	Cells, early cruise	Cells, late cruise	Surface gate
B	rigid baseline	192,588	99,111,506	73,644,516	73,673,859	pass
C	continuous TE camber	192,588	99,171,693	73,704,220	73,728,353	pass
M	continuous TE camber	192,588	99,208,470	73,727,346	73,751,122	pass
F	continuous TE camber	195,266	99,150,200	73,660,573	73,684,317	pass
W	distributed twist	195,266	99,203,781	73,668,547	73,691,820	pass
H	conventional hinged	195,266	99,121,164	73,669,659	73,693,511	pass

2.8 What was changed, in one place

Decision. The five candidates are three concepts and two selection generations. They act on the outer 40 per cent of span and move between 3.1 and 7.3 mm of trailing edge. C, F and M differ in amplitude and spanwise placement rather than in mechanism, and all three close to zero at the tip. W and H are the structurally distinct pair: W rotates whole sections bidirectionally and is the only candidate that moves a leading edge, and H breaks the surface slope at a hinge.

3 OpenFOAM setup and the automated pipeline

Purpose. Every number in this report depends on a solver recipe, a trim convention and a convergence gate, and every case reached its number down one automated route. This section fixes each of them once, so that no result section has to restate them and no two results can rest on different ones.

3.1 Solver, boundary conditions and mesh

Recipe. Every value here was read from the recipe folder each case was built from, and the recipes were confirmed against the built cases SWB.a1p60 and CMPB.a0p87. Per-geometry surface and cell counts are in Table 6.

Method. Both conditions run `foamRun` under OpenFOAM-org 12 [5], steady state, SIMPLEC. Condition CR uses the `incompressibleFluid` module at $\nu = 1.46 \times 10^{-5} \text{ m}^2/\text{s}$. Cruise uses the `fluid` module in transonic mode with a perfect gas. Table 7 gives both setups.

Two models, by design. Condition CR runs kOmegaSST wall resolved and cruise runs Spalart-Allmaras wall modelled. The criterion is the drag split: skin friction cancels between the members of a morph pair, and Spalart-Allmaras is the stronger model on the drag-rise derivative that does not. `nutLowReWallFunction` sets $\nu_t = 0$ at the wall and is the resolved-sublayer condition. Within a condition the model and the wall treatment are uniform.

Numerics. The cruise schemes and solution control are byte-copied from the project’s Onera M6 case. That case reproduces AGARD AR-138 run 308 [3]. Condition CR runs limited gradients, a limited surface-normal gradient and one non-orthogonal corrector.

Residuals. Residual targets sit below anything these solves reach. Each run therefore ends at `endTime`, and the force gate of Section 3.4 decides. Achieved residuals are flat rather than falling. Condition CR reaches $U_x \sim 10^{-9}$ and cruise $U_x \sim 1.8 \times 10^{-7}$.

Table 7: Discretisation and solution control, read from the two recipe folders and confirmed against the built cases. Residual targets are set below anything the solve reaches, so the run ends at `endTime` and the gate of Table 11 decides; the achieved plateau levels are given alongside. The cruise column serves both $M = 0.78$ states, which differ only in freestream state and layer stack.

	Condition CR	Cruise (early and late)
Module, fluid	<code>incompressibleFluid</code> , $\nu = 1.46 \times 10^{-5}$	<code>fluid</code> transonic, C_p 1005, Pr 0.71
Turbulence model	kOmegaSST	Spalart-Allmaras
ν_t wall condition	<code>nutLowReWallFunction</code>	<code>nutUSpaldingWallFunction</code>
Gradient	Gauss linear, cell limited on U and turbulence	
Convection, U	bounded linearUpwind	linearUpwind
Convection, turbulence	bounded limitedLinear 1	linearUpwind
Convection, energy	n/a	<code>filteredLinear2</code> 0.2 0
Laplacian, snGrad	limited corrected 0.33	corrected
p solver	GAMG 10^{-8} , relTol 0.01	GAMG 10^{-8} , relTol 0.1
U , turbulence solver	smoothSolver 10^{-9}	GAMG 10^{-8} , relTol 0.1
Non-orth. correctors	1	0
Relaxation	U 0.5, all 0.5	p 1, U 0.9, e 0.8
Residual targets	p 10^{-9} , U 10^{-10}	p 10^{-9} , U 10^{-10} , e 10^{-9}
Residuals achieved	$U_x \sim 10^{-9}$, $p \sim 10^{-8}$	$U_x \sim 1.8 \times 10^{-7}$, $p \sim 2 \times 10^{-8}$

Domain. The domain is free air. One background block spans $60 \times 30 \times 60 \text{ m}$, which puts the far field 16.4 semispans out. The root plane is a symmetry plane and the wing is `noSlip`. Table 8 gives the patch conditions per field. **Incidence.** Incidence enters through the freestream vector and the force axes. Condition CR runs

$U_\infty = (33.991249, 0, 0.771369)$ m/s. Early cruise runs $(233.907340, 0, 3.533537)$ m/s. Each case's α is read back from the **dragDir** header the solver wrote, never from the case name.

Turbulence inflow. Every case is fully turbulent from the leading edge. Condition CR carries $k = 1.734 \times 10^{-3} \text{ m}^2/\text{s}^2$ and $\omega = 14.25 \text{ s}^{-1}$. That is a turbulence intensity of 0.10 per cent and $\nu_t/\nu = 8.3$. Cruise carries $\tilde{\nu} = 1.3056 \times 10^{-5} \text{ m}^2/\text{s}$, and zero at the wall.

Table 8: Boundary conditions as built, per patch and field. Condition CR fixes pressure at the outlet and leaves the other far-field patches **zeroGradient**; the compressible cases use the **freestream** pair on all four far-field patches. Forces are integrated over the **wing** patch alone.

Patch	Condition CR	Cruise (early and late)
wing	U noSlip, p zeroGradient k kLowReWallFunction ω omegaWallFunction ν_t nutLowReWallFunction	U noSlip, p zeroGradient T zeroGradient, $\tilde{\nu}$ fixed 0 ν_t nutUSpaldingWallFunction α_t alphasatWallFunction
symmetry	symmetry	symmetryPlane
inlet, topBottom, outboard	U , k , ω inletOutlet p zeroGradient, ν_t calculated	U freestreamVelocity, p freestreamPressure T , $\tilde{\nu}$ inletOutlet, ν_t calculated
outlet	p fixedValue 0, U inletOutlet	as the other far-field patches

Mesh. One mesh recipe serves both conditions, and only the prism stack differs. The surface cell is set by the 1.03 mm blunt trailing edge, not by wall treatment or Reynolds number. Refinement level 11 gives a 0.488 mm surface cell and 2.11 cells across the trailing-edge base. The two recipes agree on wall-face count to three faces in 5.28×10^6 , which is the measured form of the claim that one recipe serves both conditions. Table 9 gives the recipe and what was built from it. **Layers.** Condition CR asks 11 layers at a $4.5 \mu\text{m}$ first cell and builds 9.79 of them. Early cruise asks 5 at $43.7 \mu\text{m}$ and builds 4.96 on every cruise mesh; late cruise asks 5 at $56.1 \mu\text{m}$ and builds 4.97. Achieved stack heights are 95.2, 99.5 and 99.6 per cent of the request, read from each mesh's own log; an achieved stack can only fall below the request, and every one does. Extrusion covers 97.1 per cent of wall faces at Condition CR and 99.91 to 99.93 per cent at cruise.

Wake boxes and cell cap. Both conditions carry two Trefftz wake boxes over the aft wing and near wake, at levels 7 and 6, under a 130 million cell cap. Every one of the twelve fails three to five **checkMesh** checks, drawn from skewness, face tetrahedra, small determinant and concavity, plus a small-interpolation-weight flag on C's early-cruise mesh alone; maximum non-orthogonality is 72.4° to 74.9° .

Achieved y^+ is the design value. The stack is sized to put the first-cell top at the nominal target. OpenFOAM evaluates y^+ at the cell centre, so a correct stack reads half the nominal figure. Condition CR is designed at 0.541 and 0.270, cruise at 80.3 and 40.1. Measured band medians are 0.279 to 0.302 and 37.5 to 41.9. Both conditions sit on their design value, and the cruise wall condition is blended across the buffer layer.

Table 9: Mesh recipe and as-built statistics. Only the layer stack differs between the two conditions. Surface cell size is $1.0/2^{11} = 0.488$ mm. Design y^+ is quoted at the first-cell top and at the cell centre, the centre being where OpenFOAM's **yPlus** object evaluates. Measured y^+ is the band median over the five sampled semispan stations of each solved trim case, six at Condition CR and six at early cruise, from the section cuts. The cruise column is the early-cruise mesh; late cruise differs only in its first layer. Per-geometry cell counts are in Table 6.

	Condition CR (wall resolved)	Cruise (wall modelled)
Background	$60 \times 30 \times 60$ m at $60 \times 30 \times 60$ cells, base 1.0 m	
Surface level	11, giving 0.488 mm; feature and TE levels also 11	
Distance shells	level 8 / 7 / 6 / 5 at 0.030 / 0.100 / 0.350 / 1.200 m	
Wake, cone, morph	level 7	
Quality limits	non-orthogonality 65 (relaxed 75), skewness 4	
Decomposition	192 subdomains, hierarchical	
Layers asked	11, first $4.5 \mu\text{m}$, r 1.450	5, first $43.7 \mu\text{m}$, r 1.300
Stack asked	0.586 mm	0.395 mm
Layers achieved	9.79 of 11, 95.2% of stack	4.96 of 5, 99.5% of stack
Extrusion	97.1% of wall faces	99.91% of wall faces
Wall faces	5,276,772	5,276,562 to 5,277,084
Max non-orthogonality	70.19 to 70.98	74.10 to 74.86
Design y^+ , top / centre	0.541 / 0.270	80.3 / 40.1
Measured y^+ , band median	0.279 to 0.302	37.5 to 41.9

3.2 The automated pipeline

Pipeline. Figure 3 is the route from a delivered geometry to a quotable number, and every case in this report went down it without manual dictionary editing. The shaded boxes are gates: a surface that is not closed, a mesh outside the quality limits, or a force history that has not settled stops the case there. Nothing is asserted by the step that was supposed to cause it. The run card is written from the case, the harvest is written from the run cards, and every table and figure in this report is regenerated from the harvest by script, with the checks of Section 3.5 run before any number is quoted.

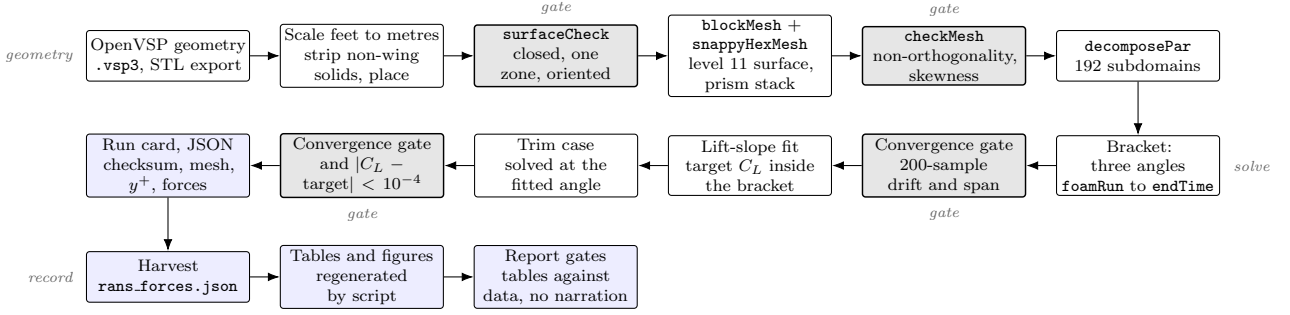


Figure 3: The automated pipeline, one case from geometry to record. Shaded boxes are gates, and a case that fails one stops there.

3.3 Trim convention: every comparison is at matched lift

Purpose. Drag is compared at the same lift, not at the same angle of attack. A morphed wing carries the target lift at a different incidence from the baseline. Comparing the two at one fixed angle would read a lift difference as a drag difference. This is the single convention the whole report rests on.

Method at Condition CR. Three angles bracketing the expected trim are solved to convergence. Three geometries trim below their first bracket and carry a fourth bracket angle, lower than the other three. C_L and C_D window means are fitted by least squares through all bracket angles, and the fit residuals are printed. The trim angle follows from that fit, and a separate trim case is solved at it: the fourth case run for most geometries, and the fifth for the three that needed an extra bracket. That solved case is the quotable point, not an interpolation. The target must lie inside the bracket, and an extrapolated angle is refused.

Two details that keep the convention honest. Incidence is read from each case’s own freestream vector, not from its directory name; CMPC_aOp87 actually runs 0.8655 deg. All three bracket cases are asserted to share one operating point, derived from each case’s own density and agreed to 10^{-12} , before any fit is formed.

Method at cruise. The cruise trims were reached by continuation rather than by bracketing. Each case converged a first leg at an initial incidence, was nudged in α on its own lift slope (a prior slope for the first nudge, the secant through its own legs after that), and continued until $|\Delta C_L| < 10^{-4}$. Each nudge rotates the freestream and the force-coefficient axes together, and the solved axes are asserted to agree with the trimmed incidence to 10^{-6} deg. Nine of the twelve needed one nudge and three needed two; Table 10 lists every leg. The off-trim legs converged on the same gate as the trims and are the only points the cruise drag-due-to-lift slopes use.

Table 10: Every cruise trim leg. $\Delta\alpha$ is the nudge applied after the leg. LCF’s third leg stopped after one iteration and was continued as its fourth, so its legs read 1, 2, 4. Source: `data/rans_forces.json`, `trim_history` of each case.

Geom.	Leg	Iter.	α (deg)	C_L	C_L -tgt (ct)	C_D (ct)	$\Delta\alpha$ (deg)
<i>Early cruise, target $C_L = 0.529297087450$</i>							
B	1	2000	1.096175	0.5284758	−8.213	177.569	+0.005905
	2	4000	1.102080	0.5293165	+0.194	177.839	trimmed
C	1	2000	0.738736	0.5289479	−3.492	190.685	+0.002566
	2	4000	0.741302	0.5292666	−0.305	190.710	trimmed
F	1	2000	0.935849	0.5399245	+106.274	183.659	−0.078137
	2	4000	0.857712	0.5291656	−1.315	179.584	+0.000955
	3	6000	0.858667	0.5292954	−0.017	179.635	trimmed
H	1	2000	1.007185	0.5392190	+99.219	183.782	−0.072354
	2	4000	0.934831	0.5293058	+0.087	180.083	trimmed
M	1	2000	0.747956	0.5290163	−2.808	192.302	+0.002064
	2	4000	0.750020	0.5293351	+0.380	192.393	trimmed
W	1	4000	0.964951	0.5389152	+96.181	180.750	−0.069505
	2	6000	0.895446	0.5292406	−0.565	177.070	trimmed
<i>Late cruise, target $C_L = 0.544114800210$</i>							
B	1	2000	1.258829	0.5434626	−6.522	187.722	+0.004720
	2	4000	1.263549	0.5441685	+0.537	187.959	trimmed
C	1	2000	0.905084	0.5436205	−4.943	200.023	+0.003644
	2	4000	0.908728	0.5441900	+0.752	200.173	trimmed
F	1	2000	1.102702	0.5552063	+110.915	193.696	−0.082086
	2	4000	1.020616	0.5439620	−1.528	189.246	+0.001115
H	4	6000	1.021731	0.5440928	−0.220	189.307	trimmed
	1	2000	1.173633	0.5544600	+103.452	194.032	−0.075956
	2	4000	1.097677	0.5439945	−1.203	189.847	+0.000873
M	3	6000	1.098550	0.5440776	−0.372	189.886	trimmed
	1	2000	0.914829	0.5437200	−3.948	201.590	+0.002907
W	2	4000	0.917736	0.5441693	+0.545	201.687	trimmed
	1	2000	1.129025	0.5540568	+99.420	191.246	−0.072295
	2	4000	1.056730	0.5440253	−0.895	187.178	trimmed

Result. The trim outcome for every geometry and condition is in the performance tables, Tables 14, 15 and 16, which carry each trim’s incidence and its lift residual against that condition’s own target; Figure 4 is the picture of the Condition CR procedure. The project tolerance is $|\Delta C_L| < 1 \times 10^{-4}$, one count of lift, and all eighteen trims are inside it: the widest is `LCW_trim` at 0.895 counts. Each family’s own drag-due-to-lift slope converts a lift residual into drag, and the largest such bias at any condition is 0.036 counts, on `LCW_trim`. At cruise six of the twelve slopes rest on a nudge too small to resolve the slope, under 0.002 in C_L ; they are flagged in the data file as fit to the trim residual only, and they bound it well below 0.1 counts either way. The same conversion over the six Condition CR trims gives at most 0.003 counts.

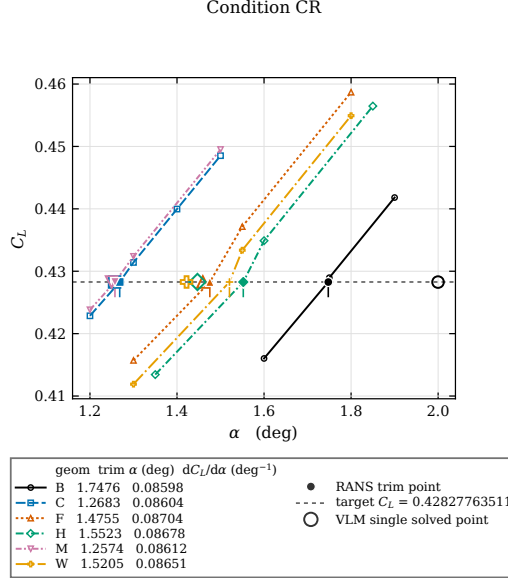


Figure 4: The trim procedure made visible at Condition CR; the cruise trims were reached by continuation and are tabulated in Table 10. C_L against α over each family’s solved angles, with the target drawn as a dashed line and each solved trim case filled. Rings are single-point VSPAERO trims, drawn to show agreement on incidence and carrying no lift slope of their own. DSO basis, half model, $A_{\text{ref}} = 0.620462 \text{ m}^2$.

Decision. Every drag count in this report is a difference taken at matched lift, inside one condition, on one reference frame. The residuals above are worth hundredths of a count of drag at most, against candidate differences of tenths of a count at Condition CR and up to fifteen counts at cruise. The comparison is therefore read as a drag comparison at fixed lift, with no lift correction applied to any number.

3.4 Convergence gate

Method. A steady SIMPLE solve on this wing limit-cycles about a stationary mean. The gate is therefore written on the drift of the mean and on the span across the window. It is evaluated once at the end of each leg, over the last 200 force samples. All three terms must pass, and the verdict is re-derived from the forces by a second script. Table 11 gives the thresholds and what the 55 cases converged at the 15 September regeneration achieved. **That set is not the 90 converged cases counted in Table 12;** the rest landed after it. The gate itself is unchanged and every later case passed it on its own force history. Over the 27 cruise trims and legs that carry every cruise result in this report, the achieved values are C_d drift 0.0001 to 0.0240 ct (median 0.0022), C_l drift 0.0037 to 0.4007 ct (median 0.0559) and C_l span 0.0013 to 0.8637 ct (median 0.1223), all 27 CONVERGED over 200 samples; the widest are first legs, before the nudge, and every trim sits well inside. What is not claimed is that the tabulated statistics were recomputed over all ninety.

Sampling. 200 samples is a different number of iterations per condition. Condition CR writes forces every fifth time step and cruise every step. On a leg ending at `endTime` 4000, which is 46 of those 55, the windows therefore span $t = 3005$ to 4000 and $t = 3801$ to 4000. They shift with `endTime` on the other nine, whose values are given under Run length below; the window is always the last 200 samples, never a fixed t .

Reported value. Every force quoted in this report is a 200-sample window mean. 43 markers carry the gate’s own window means. The remaining 12 are recomputed from the merged force history over the same window. That route reproduces the 43 window-mean markers to 5.0×10^{-4} counts in both C_d and C_l .

Run length. `endTime` takes three values across the campaign: 46 cases at 4000, 7 at 3000 and 2 at 4500. Every converged case reached its `endTime`.

3.5 What the evidence covers

Coverage. Ninety cases have converged: 25 at Condition CR, 33 at early cruise and 32 at late cruise. The campaign is complete; every geometry is solved at every operating point. Each passed the same gate on its own force history, over a 200-sample window. Mean drift stays below 0.05 counts in C_D and 1.0 count in C_L . The C_L span across the window stays below 1.0 count. Every quoted force is that window mean.

Table 11: The convergence gate and the margins achieved over the 55 cases converged at the 15 September regeneration. Drift is the difference of the two half-window means and span is the first-to-last excursion across the window. Drag in counts on the DSO basis, half model, $A_{\text{ref}} = 0.620462 \text{ m}^2$; C_l terms in counts of C_L . Values are each case’s own marker, harvested 2026-09-15T14:05:48Z from `data/rans_forces_current.json`.

Term	Threshold	Achieved over the 55	Median
Window W	200 samples	200 on all 55	200
C_d drift	≤ 0.05 ct	0.0000 to 0.0130	0.0012
C_l drift	≤ 1.0 ct	0.0004 to 0.2590	0.0615
C_l span	≤ 1.0 ct	0.0008 to 0.5374	0.1068
Verdict	all three terms	CONVERGED, 55 of 55	n/a

What the cruise results rest on. Of the 65 cruise cases, 27 carry every cruise number in this report: the twelve trims and the fifteen off-trim legs that reached them (Table 10). The other 38 are alpha brackets kept in the data file and used by no cruise result.

Table 12 is the deliverable: six geometries, three operating points, and the solved cases behind each.

Result. All three conditions are complete at six trims of six. Trimmed lift lands within 0.10 counts of target at Condition CR, within 0.57 counts at early cruise and within 0.90 counts at late cruise.

Table 12: Converged cases by geometry and operating point. The three points are Condition CR ($M = 0.10$, $C_L = 0.428277635108$), early cruise ($M = 0.78$, $C_L = 0.529297087450$) and late cruise ($M = 0.78$, $C_L = 0.544114800210$). Every column carries its geometry’s solved trim. At cruise, “trim and legs” counts the trim and its off-trim legs and “brackets” the alpha brackets no cruise result uses. Source: `data/rans_forces.json`, rows generated by `scripts/make_cruise_tables.py`.

ID	Concept	Condition CR	Early cruise		Late cruise	
		solved	brackets	trim and legs	brackets	trim and legs
B	rigid baseline	4	3	2	3	2
C	continuous TE camber	5	4	2	3	2
F	continuous TE camber	4	3	3	3	3
M	continuous TE camber	4	4	2	3	2
W	distributed twist	4	3	2	3	2
H	conventional hinged	4	3	2	3	3
Totals		25	20	13	18	14

Two data files, and why both names appear. `data/rans_forces_current.json` is the raw harvest: one record per case, read off each case’s own marker and `forceCoeffs` output. `data/rans_forces.json` is the structured merge of that harvest with a per-case provenance pass, and it is the file that carries the frame assertions, the trim deltas and the coverage partitions. The performance tables cite the harvest because that is what they were built from; Figure 5 cites the merge because that is what it reads. The two agree on every value quoted in those tables, and that is checked rather than assumed: the figure writes every plotted number to `fig/fig_trimmed_performance_values.csv` at full precision, and all 90 comparisons against the three performance tables, 18 rows by α , C_D , ΔC_D , C_L -tgt and L/D , agree to the precision printed.

4 Quantities of interest

Purpose. This section is the contents of the results that follow, as a table. It names every quantity the campaign reports, the frame each one carries, how much of the campaign it covers, and whether the low-order study produces it at all, which is what decides where a comparison can be made and where the RANS stands alone.

4.1 Quantities and their frames

Frame. Every coefficient here is on the DSO basis. The RANS cases solve a half wing, so $A_{\text{ref}} = 0.620462 \text{ m}^2$ and $l_{\text{ref}} = 0.393957 \text{ m}$. Twice A_{ref} matches the DSO S_{ref} to 3.2×10^{-6} relative. All 90 converged cases carry that one frame, uniform within each condition. Differences are formed inside a single condition, never across two.

Definitions. Table 13 is the register. One drag count is 10^{-4} in a force coefficient. Coverage is stated per quantity, because the force set and the surface set were harvested on different days: forces cover 90 cases and 18 trims, surface data all 18 trims. The last two columns say whether the low-order study produces the quantity, and where in this report it is compared.

Drag split. The split comes from a `forces` object on the same drag direction and the same q as the coefficients. The test is closure against the solver’s own C_D . On `SWB_a1p90` the parts give $91.218 + 104.098 = 195.316$ ct against 195.317 ct. At cruise the closure holds once q carries p : `CMPB_a1p20` gives $131.534 + 51.917 = 183.451$ ct against 183.451 ct.

Wall quantities. c_p is formed inside the solver, so its normalisation travels with the file. That q is $578 \text{ m}^2/\text{s}^2$ at Condition CR, 11292.8 Pa at early cruise and 8257.37 Pa at late cruise, each verified against its own independently known value. Skin friction is carried twice, as a magnitude for level and as a signed freestream-aligned component for direction. The sign multiplier is measured on inboard mid-chord faces whose attachment is known, and came out 100 per cent one sign on all eighteen trims. Separation is therefore read

Table 13: Quantities of interest, their definitions and their frames. DSO basis throughout: “DSO half” is $A_{\text{ref}} = 0.620462 \text{ m}^2$ with $l_{\text{ref}} = 0.393957 \text{ m}$, half of $S_{\text{ref}} = 1.24092 \text{ m}^2$ to 3.2×10^{-6} . q is the freestream dynamic pressure, kinematic at Condition CR and absolute at cruise. Surface quantities carry no reference area. “VLM” says what the low-order study produces for the quantity and “Compared” whether this report sets the two side by side. Source: `data/rans_forces_current.json` and `data/qoi_methods.md`.

Quantity	Definition	Frame	RANS coverage	VLM	Compared	Section
C_L, C_D	forceCoeffs on the wing patch	DSO half	90 cases	$C_{D_{\text{tot}}}$, not compared	no	5.1, 5.2
ΔC_D	candidate minus baseline at fixed C_L	DSO half	18 trims	induced only	ordering	6.2
C_M	forceCoeffs about (1.34, 0, 0) m	DSO half, l_{ref}	per case	other frame	no	
$C_{D,p}, C_{D,v}$	forces , same $\hat{d}$ and same q	DSO half	both conditions	no	no	4.1
L/D	C_L/C_D at the trim point	area cancels	18 trims	no	no	5.1, 5.2
Trim α	bracket, slope fit, then a solved case	solver dragDir	18 trims	yes	yes	5.3
c_p	$(p - p_\infty)/q$, formed in the solver	q per condition	90 bands	no	no	A
$c_{f,s}, c_{f,s}$	$ \tau_w /q$, and its signed form	per-file q	90 bands	no	no	A
y^+	yPlus object and surface field	cell centre	per case	no	no	3.1
Spanwise load	c_l/C_{ref} , pressure only	C_{ref} DSO	18 trims, 78 stations	69 strips	yes	5.4
$C_{D,i}, e$	Trefftz integral of $(v'^2 + w'^2)$	half-wing A_{ref}	4 planes, 6 trims	near and far	yes	5.5
M_h	hinge moment about $x_h/c = 0.62$	local chord	18 trims	not computed	no	5.6
M_x	root bending, solver moment at $y = 0$	N m	18 trims	yes	yes	5.7

from a sign rather than a threshold. Median upper-surface y^+ is 0.28 to 0.30 at Condition CR and 37 to 42 at both cruise points.

Spanwise load and induced drag. Sectional load is taken from the wing surface at 78 stations, pressure only. The test is again a closure: the strips integrate back to the case’s own C_L as $C_L = 2 \int c_l c dy / S_{\text{ref}}$, with the explicit half-wing factor of two. That closure reads 4.65 to 4.91 per cent low across the six Condition CR trims and 4.85 to 5.14 per cent low across the twelve cruise trims, the gap being the root and tip outside the sampled window $\eta = 0.025$ to 0.986. Induced drag is defined as a Trefftz integral on four planes per case at $x_{TE} + fc$, $f = 0.05$ to 0.75. The induced drag those planes give, and the span efficiency it implies, are reported in Section 5.5.

5 Results, quantity by quantity

Purpose. Each quantity of Table 13 in turn, with the RANS result first and the low-order study’s value beside it wherever the study produces one. Every coefficient is on the DSO basis and every difference is formed inside one operating point; the comparison of the two methods as a selection tool is drawn together in Section 6.

5.1 Trimmed drag at Condition CR

Result. Table 14 gives the low-speed result, and the headline is how little separates the candidates. The five span 1.1 counts, from W at -0.868 to M at $+0.232$. Four of the five sit below the baseline and one above it, but every one of them is inside a count and a half. On L/D the whole field lies between 22.75 and 22.88, a spread of an eighth of a point; W reaches 22.882 against the baseline’s 22.776. Every candidate reaches target lift at lower incidence than the baseline, by 0.19 to 0.50 deg: the aft camber does its intended job at this condition.

Ordering. The concept ordering differs between the two points, and at low speed only the grouping is resolvable, not the sequence within it. At Condition CR W and F are the two cheapest, separated by 0.053 counts, with F sitting 0.806 counts below H; at early cruise C and M are the two cheapest and F is the most expensive of all five. Differences of a twentieth of a count between adjacent pairs at low speed are the size of the convergence gate itself, so the low-speed sequence is read as a group rather than a ranking, even though the 1.1-count spread across the whole field is twenty-two times that gate and is real. The cruise points, where the field spans forty counts, are where the ordering carries. Note also that C, M and F are all continuous trailing-edge camber and do not group together at either point: the concept does not by itself decide the cost.

Table 14: Trimmed performance at Condition CR, $M = 0.10$, target $C_L = 0.428277635108$, kOmegaSST wall resolved. DSO basis, half model, $A_{\text{ref}} = 0.620462 \text{ m}^2$; one drag count is 10^{-4} . ΔC_D is against **SWB_trim** at the same condition and the same target lift, negative being a saving. $C_L\text{-tgt}$ is in counts of C_L . Rank is by ΔC_D , cheapest first. Source: `data/rans_forces.json`, rows emitted by `scripts/make_cr_table.py`.

Geom.	Case	α (deg)	C_D (ct)	ΔC_D (ct)	$C_L\text{-tgt}$ (ct)	L/D	Rank
B	SWB_trim	1.7476	188.031	+0.000	−0.094	22.776	−
C	SWC_trim	1.2683	187.840	−0.191	−0.005	22.800	3
M	SWM_trim	1.2574	188.263	+0.232	+0.001	22.749	5
F	SWF_trim	1.4755	187.216	−0.815	−0.039	22.876	2
W	SWW_trim	1.5205	187.163	−0.868	−0.037	22.882	1
H	SWH_trim	1.5523	188.022	−0.009	−0.039	22.778	4

5.2 Trimmed drag at cruise

Purpose. The two cruise points are early cruise at $M = 0.78$ and $C_L = 0.529297087450$ and late cruise at $M = 0.78$ and $C_L = 0.544114800210$. The two are separate operating points: each carries its own baseline trim, every increment is formed inside one of them, and nothing is differenced across them or against Condition CR.

Both run Spalart–Allmaras wall modelled where Condition CR runs kOmegaSST wall resolved, so counts from the two regimes are set beside one another as magnitudes only.

5.2.1 Early cruise

Result. Table 15 gives the cruise result, and it is the decision-level one. **W is the only candidate that saves drag**, by 0.769 counts against a baseline of 177.839, which raises L/D from 29.764 to 29.889. F and H cost 1.796 and 2.244 counts, 0.30 and 0.37 points of L/D . C and M cost 12.871 and 14.554 counts, 2.0 and 2.3 points of L/D . Every trim lands within 0.565 ct of target C_L , inside the one-count tolerance; the drag bias that residual implies is at most 0.022 ct and is carried per case in the data file.

Incidence. Each candidate again trims at lower incidence than the baseline, by 0.17 to 0.36 deg. The commanded camber delivers its lift at cruise as designed. The drag consequence depends on where it puts the recompression (Section 5.2.4).

Table 15: Trimmed performance at early cruise, $M = 0.78$, target $C_L = 0.529297087450$, Spalart–Allmaras wall modelled. Frame and conventions as Table 14, with the baseline `CMPB.trim`; rows ranked by ΔC_D . Source: `data/rans_forces.json`, rows generated by `scripts/make_cruise_tables.py`.

Geom.	Case	α (deg)	C_D (ct)	ΔC_D (ct)	C_L -tgt (ct)	L/D	Rank
B	<code>CMPB.trim</code>	1.1021	177.839	+0.000	+0.194	29.764	–
W	<code>CMPW.trim</code>	0.8954	177.070	–0.769	–0.565	29.889	1
F	<code>CMPF.trim</code>	0.8587	179.635	+1.796	–0.017	29.465	2
H	<code>CMPH.trim</code>	0.9348	180.083	+2.244	+0.087	29.392	3
C	<code>CMPC.trim</code>	0.7413	190.710	+12.871	–0.305	27.752	4
M	<code>CMPM.trim</code>	0.7500	192.393	+14.554	+0.380	27.513	5

5.2.2 Late cruise

Result. Table 16 gives the late-cruise result, and it reproduces the early-cruise picture. W again saves drag, by 0.781 counts against a baseline of 187.959; F and H cost 1.348 and 1.927; C and M cost 12.214 and 13.728. Every trim lands within 0.895 ct of target C_L , the widest being `LCW.trim`, and incidence again falls below the baseline’s, by 0.16 to 0.35 deg.

Table 16: Trimmed performance at late cruise, $M = 0.78$, target $C_L = 0.544114800210$, Spalart–Allmaras wall modelled. Frame and conventions as Table 15, with the baseline `LCB.trim`. Same turbulence model and wall treatment as early cruise, so these counts are directly comparable with Table 15. Source as Table 15.

Geom.	Case	α (deg)	C_D (ct)	ΔC_D (ct)	C_L -tgt (ct)	L/D	Rank
B	<code>LCB.trim</code>	1.2635	187.959	+0.000	+0.537	28.951	–
W	<code>LCW.trim</code>	1.0567	187.178	–0.781	–0.895	29.065	1
F	<code>LCF.trim</code>	1.0217	189.307	+1.348	–0.220	28.741	2
H	<code>LCH.trim</code>	1.0986	189.886	+1.927	–0.372	28.653	3
C	<code>LCC.trim</code>	0.9087	200.173	+12.214	+0.752	27.186	4
M	<code>LCM.trim</code>	0.9177	201.687	+13.728	+0.545	26.981	5

5.2.3 The three conditions together

Result. The two cruise points agree with each other exactly, and they agree with Condition CR on everything the low-speed field can resolve. Between early and late cruise *zero* of the ten candidate pairs change order: both give W, F, H, C, M. At Condition CR the sequence is W, F, C, H, M (Table 14). The one pair that differs is C and H, which at Condition CR lie 0.182 counts apart, inside the resolution Section 5.1 gives neighbours at low speed. **W is the cheapest geometry at all three operating points**, at –0.868, –0.769 and –0.781 counts. What changes is the magnitude: the field spans 1.1 counts at Condition CR and 15.3 at early cruise, almost all of it carried by C and M.

What the third condition establishes, which two could not. A single agreement between two conditions could be a property of the particular pair of points chosen. A third point that reproduces the second exactly is evidence that the cruise ordering is a property of the flow regime rather than of one operating point.

5.2.4 Why the cruise penalty appears, and why W escapes it

Mechanism. At $M = 0.78$ the upper surface at $\eta = 0.80$ is supersonic over most of the chord on every geometry, 144 to 153 of 200 chordwise bins below $c_p^* = -0.4931$ at early cruise. What separates the candidates is how that supersonic region ends. On B and W it recompresses without a shock: the steepest gradient is $dc_p/d(x/c) = 4.22$ and 4.53, both at $x/c = 0.8725$, below the shock threshold of 15. F and H end it in a moderate shock, 17.1 at $x/c = 0.7525$ and 19.1 at 0.7725, and stay attached. C and M end it in a strong shock, 45.7 at 0.7375 and 50.7 at 0.7275, and separate behind it. **The drag ordering follows the shock strength:** W within a count of the baseline, F and H about two counts above it, C and M twelve to fifteen. Late cruise repeats the pattern at slightly lower strength, 16.1 and 16.8 on F and H and 43.6 and 38.0 on C and M. At $M = 0.10$ the same camber meets subsonic flow and the field stays inside a count and a half.

Where the load moves. Figures 6 and 7 show the redistribution. Every candidate unloads the inboard wing and loads the outboard wing; the difference from B crosses zero between $\eta = 0.571$ and 0.652 and peaks between 0.725 and 0.800 , at the same stations to three decimals at both cruise points. The inboard and outboard integrals cancel, as they must at fixed lift. C and M carry the largest outboard increment, $+0.075$ and $+0.077$ in $c_l c / C_{\text{ref}}$ at early cruise, against $+0.046$ for F, H and W, and that peak sits under the outboard shock.

Separation. Two bands separate in the 30 sampled at each cruise point, and they are the same two: CMPC.trim and CMPM.trim at $\eta = 0.80$ (Figure 8), and their late-cruise counterparts. Reversed flow covers 23.3 per cent of C’s upper-surface faces in that band at early cruise, from $x/c = 0.743$ to 0.992 , and 23.9 per cent of M’s, from 0.730 to 0.997 ; at late cruise the figures are 24.1 and 25.3 per cent. Every other cruise band reverses at most 0.019 per cent of its upper-surface faces, and nothing separates at Condition CR. The cases are wall modelled, so the sign of this result is firm and its extent carries the precision of the section cuts. The strong shock and the separation occur together, on the two strong-camber candidates, and they are the two most expensive. Appendix A.3 gives the station detail.

Section pressure and skin friction at the five spanwise stations, at Condition CR and both cruise points, are in Appendix A.

5.3 Trim angle

Result. Trim angle is where the two methods agree best (Table 17, Figure 14b). VSPAERO matches the solved RANS angle to within 0.105° on all five candidates, and to 0.016° or better on three of them. Carried through each family’s own fitted lift slope, that is a lift error of at most 2.13 per cent of the target C_L . The baseline B is the outlier at $+0.252^\circ$, and its deck is the only Condition CR run to miss its own target, by 1.53×10^{-5} in C_L .

Slope. The RANS side fits $dC_L/d\alpha = 0.0860$ to 0.0870 per degree over 4 or 5 solved angles per family, with residuals of 1.6×10^{-4} to 1.1×10^{-3} in C_l . The decks are single-point, so the VLM side carries no lift slope and the comparison is made on trim angle, which needs no derivative.

Table 17: Trim angle at the common Condition CR target $C_L = 0.428277635108$, both sides on the DSO basis, conversion factor exactly 1.000000. RANS α comes from the solved trim case’s own `dragDir` header. Implied ΔC_L is $\Delta\alpha$ through that family’s own fitted RANS slope, as a percentage of the target. **Each slope is fitted over that family’s published alpha-sweep legs together with its Condition CR trim case, and the trim case alone was solved again on the rebuilt mesh: each slope is therefore fitted over legs from two different meshes and is not a single-mesh derivative.** They enter only $\Delta\alpha$ converted to an implied ΔC_L , which is at most 2.13 per cent of target, and no drag number in this report is formed through them. Rows generated by `scripts/make_rank_table.py` from `fig/fig_vlmerr_values.json`.

Geom.	α_{RANS} [deg]	α_{VLM} [deg]	$\Delta\alpha$ [deg]	implied ΔC_L [%]	$dC_L/d\alpha$ [/deg]
B	1.747609	1.99981	+0.2522	+5.06	0.085980
C	1.268311	1.25652	−0.0118	−0.24	0.086042
M	1.257423	1.24972	−0.0077	−0.15	0.086119
F	1.475456	1.45936	−0.0161	−0.33	0.087035
W	1.520545	1.42193	−0.0986	−1.99	0.086510
H	1.552310	1.44697	−0.1053	−2.13	0.086784

5.4 Spanwise load

Result. Figure 9 shows the sectional load defined in Section 4.1 for every geometry at its trim, with the VSPAERO loading beside it. The absolute curves lie on top of one another and show the planform. The difference from baseline isolates the morphing, and it sits over the morphed span in every case; how closely the two methods agree on that redistribution is Section 5.4.

Result. The VLM puts the load where the RANS puts it (Figure 9). Each side is differenced against its own baseline, which is the quantity the optimiser acts on. Over 78 stations the difference between the two redistributions has an r.m.s. of 0.0015 to 0.0037 in $c_l c / C_{\text{ref}}$, against RANS peaks of 0.034 to 0.080. That is 4.5 to 9.5 per cent of the peak. The station of peak change agrees to within 0.049 in η , and to within 0.014 on four of the five candidates. **Resolution.** The RANS gives 78 stations and the VLM 69 strips, interpolated onto a common grid with an error bound of 8.5×10^{-4} . The VLM strips close to their own polar C_L to 4.6×10^{-6} once the half-wing factor of two is applied, and to 50.0 per cent without it. The RANS loading is integrated from the wing surface, pressure only; the VLM loading comes from the 69 strips of each deck’s own `.lod`. Both are on $C_{\text{ref}} = 0.39396$ m, the DSO basis.

5.5 Induced drag and span efficiency

Method. Induced drag is taken from the crossflow kinetic energy crossing four transverse planes in the wake, at $x = 2.394, 2.626, 2.916$ and 3.206 m, sampled by a function object inside the solver rather than reconstructed afterwards. Span efficiency follows from it as

$$e = \frac{C_L^2}{\pi AR C_{Di}}, \quad (3)$$

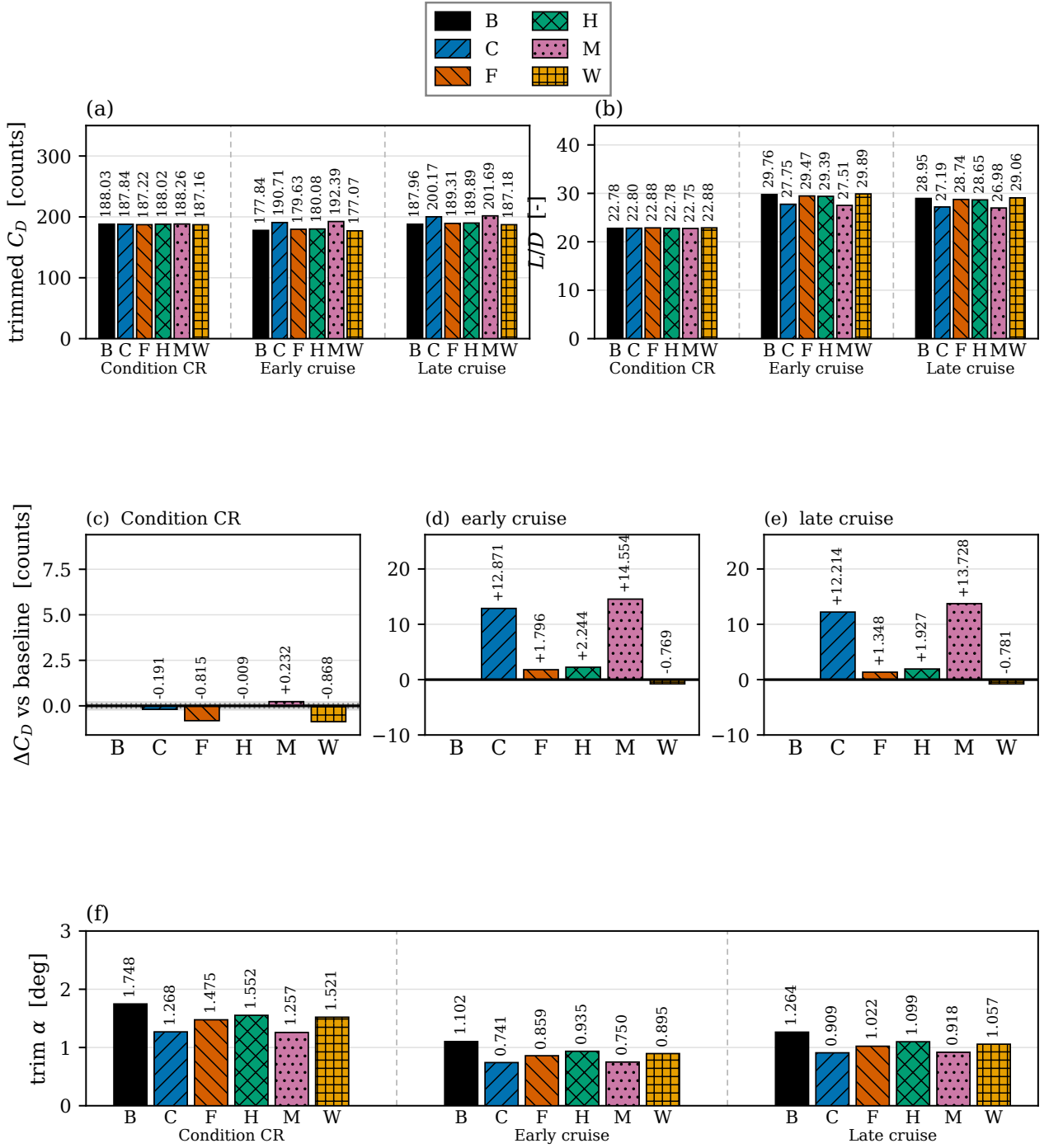


Figure 5: Trimmed performance at matched lift, all six geometries, all three conditions. Panels: (a) trimmed C_D , (b) L/D , (c) ΔC_D at Condition CR, (d) ΔC_D at early cruise, (e) ΔC_D at late cruise, and (f) trim incidence. (d) and (e) share one vertical scale; Condition CR keeps its own or its deltas would be invisible. Every delta is formed inside one condition against that condition's own B trim. DSO basis, half model, $A_{\text{ref}} = 0.620462 \text{ m}^2$. Source: `data/rans_forces.json`.

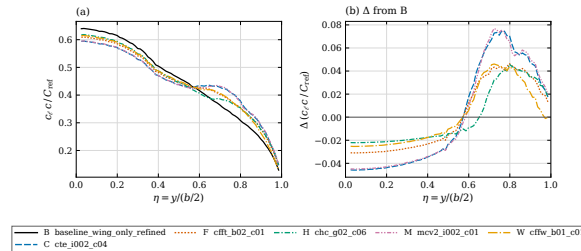


Figure 6: Spanwise loading at early cruise, $M = 0.78$, every case trimmed to $C_L = 0.529297087450$. Sectional load $c_l c / C_{\text{ref}}$ against η , and the difference from `CMPB_trim`, over 78 stations. Sectional c_l is pressure only. DSO basis, $C_{\text{ref}} = 0.39396 \text{ m}$.

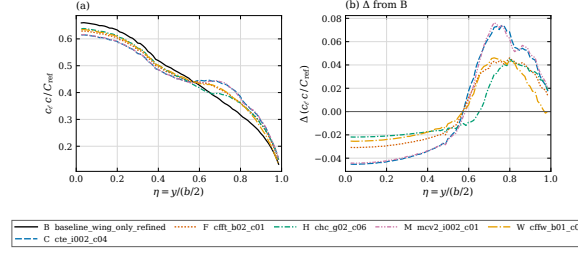


Figure 7: Spanwise loading at late cruise, $M = 0.78$, $C_L = 0.544114800210$, against LCB.trim. As Figure 6.

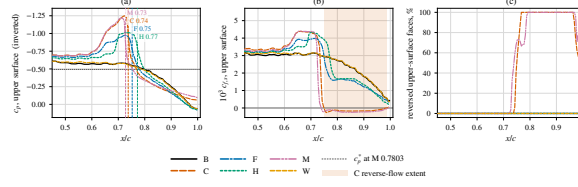


Figure 8: One of the two separated bands: CMPC.trim at early cruise, $\eta = 0.80$. Upper-surface c_p , signed freestream-aligned skin friction and reverse-flow fraction. The shock sits at $x/c = 0.7375$ with $\Delta c_p = 0.724$ and $c_p = -1.154$ upstream, against $c_p^* = -0.4931$. CPM.trim separates at the same station and is drawn beside C in Figure 17. c_p and c_f carry no reference area.

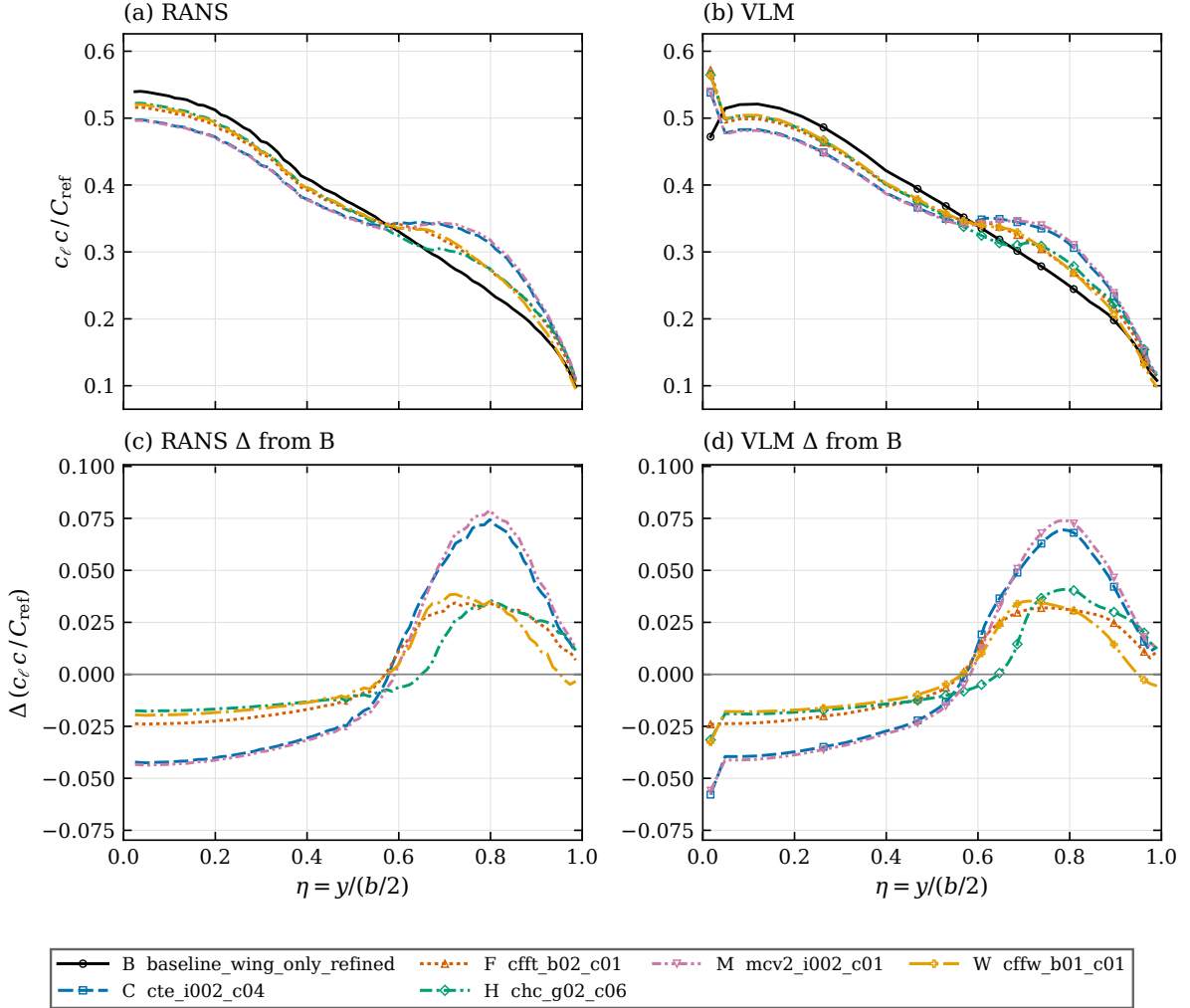


Figure 9: Spanwise loading at Condition CR, $M = 0.10$, every geometry trimmed to $C_L = 0.428277635108$ on the DSO basis. The upper row is the sectional load $c_l c / C_{ref}$ against η , where the six geometries differ by a few per cent and the curves overlaid. The lower row is each candidate minus the baseline of the same method, which is where the morphing appears. The RANS difference peaks at $\eta = 0.7985$ for C, F, H and M, and at $\eta = 0.7237$ for W, the one candidate whose command peaks furthest inboard. Peak differences are 0.079 (M), 0.075 (C), 0.040 (W), 0.037 (H) and 0.036 (F). RANS curves carry 78 dense stations over $\eta = 0.025$ to 0.986; VSPAERO curves carry 69 strips. Generated by fig/make_spanwise.py from data/spanwise/ and data/vlm.json.

with AR the full-wing aspect ratio on the DSO basis. Every geometry is evaluated at its own trimmed angle of attack, and the freestream component is removed using that case's own α ; using a different one would leave a uniform residual across the whole plane.

Frame. C_{Di} is on the half wing, $A_{\text{ref}} = 0.620462 \text{ m}^2$, matching each case's own force coefficients. e is on the full-wing aspect ratio from the DSO basis, $b_{\text{ref}} = 3.6576 \text{ m}$ and $S_{\text{ref}} = 1.24092 \text{ m}^2$. Condition CR, $U_{\infty} = 34.0 \text{ m/s}$, every geometry trimmed to $C_L = 0.428277635108$.

There is no plateau, so the spread is the uncertainty. Crossflow kinetic energy is conserved in an inviscid wake, so a Trefftz integral should not decay downstream at all. What decay remains is numerical: the grid dissipating the trailing vortex system between one plane and the next. Quoting a single station would therefore report a number without the quantity that qualifies it, and Figure 10 plots all four.

The limit at $e = 1$ is a check, not a target. For a planar wake $e = 1$ is the elliptical optimum and cannot be exceeded. That makes it useful twice over: as the reference a spanload is measured against, and as a check on the calculation itself. A value above 1 would report less induced drag than the theoretical minimum, which is not a marginal result to be discussed but an arithmetic impossibility, so any geometry returning one is withheld rather than plotted. Every value in Figure 10 sits below the line, as it must.

Result. Span efficiency at the last station ranges from 0.9673 on the rigid baseline B to 0.9859 on F, every value below the planar-wake limit. Induced drag at that station spans 54.93 to 55.98 counts.

Decay between the first and last plane is 9.5 to 10.2 per cent, and is consistent across the geometries: B 9.46 %, W 9.68 %, F 9.95 %, H 9.98 %, C 10.19 %, M 10.22 %. That consistency matters more than the magnitude, because a decay rate that differed between geometries would mean the comparison was being made on wakes resolved to different degrees, and no difference between them could be attributed to the wing.

Against the baseline. Every candidate measured here carries a higher span efficiency than the rigid baseline's 0.9673 (C, F, H, M and W). The margin is small, one to two per cent, which is the size of spanload improvement the low-order design study predicted for this class of change.

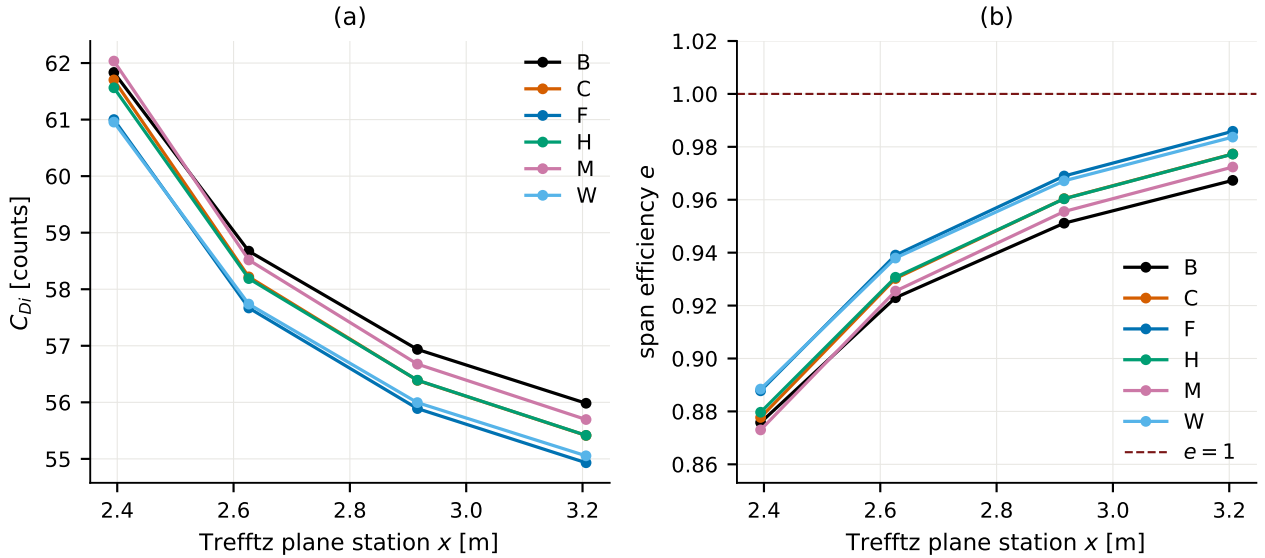


Figure 10: Trefftz-plane induced drag and the span efficiency it implies, Condition CR, $M = 0.10$, every geometry trimmed to $C_L = 0.428277635108$. Panel (a) is C_{Di} at each of the four sampled stations; panel (b) is e from Equation 3, with the planar-wake limit $e = 1$ marked. C_{Di} is on the half wing, $A_{\text{ref}} = 0.620462 \text{ m}^2$; e is on the DSO-basis full-wing aspect ratio. Generated by `fig/make_spaneff.py` from `data/span_efficiency.json`, which is assembled by `scripts/build_span_efficiency.py`, which refuses to mix meshes, to mix untrimmed legs with trimmed ones, or to carry a value above the physical limit.

Span efficiency. The two methods agree on the level once the far-field definition is used. VSPAERO reports $E_w = 0.960$ to 0.972 on the five candidates against its own near-field $E = 0.568$ to 0.587 , and the RANS Trefftz integral returns 0.972 to 0.986 on the five candidates. The near-field E is not a competing estimate of the same quantity; it is the one that produced the sign violation reported in Section 6.1.

5.6 Hinge moment about the morph line

Purpose. Two structural quantities bear on the concept selection. The hinge moment about the morph line is the load the actuator reacts, and the low-order method cannot produce it. Root bending is the active screen the design study selected on, and the low-order method does produce it, so it is the one structural quantity where the two methods compare directly. Both are reported here for every geometry at every operating point that carries them.

Purpose. This section reports the aerodynamic moment the actuator has to react. It is the one quantity in this campaign that the low-order method cannot produce at all: the DSO’s own per-case summary records `hinge_moment_status` as `not_computed_requires_surface_pressure_integration` for every one of its 23 runs, so there is no VLM column to compare against and none is constructed here. The decks do carry a C_{My} , but that is the whole-model pitching moment about the deck’s own reference point, a different quantity in a different frame, and reaching a hinge moment from it needs a transfer formula this project refuses (`scripts/hinge_moment.py`).

Definition, as agreed with the DSO team. Every element below is the agreed form and none of it is a local choice.

1. **Axis:** the swept spanwise line through $x_h/c = 0.62$ at each local section, with the moment projected on that line’s *local tangent*. The line is swept and tapered, so its tangent is not the y axis and projecting on y instead is the error a careful reader makes first.
2. **Sign:** positive is trailing-edge down. The actuator reaction is the opposite sign.
3. **Scope:** surfaces aft of the line only, over $0.60 \leq \eta \leq 1.00$, half wing. Aft-of-line is tested per station against x_h at each face’s own y , never against a single global x cut, which on a swept wing would include forward surface at the root and exclude aft surface at the tip.
4. **Sectional:** m'_h in N m/m and $c_{mh} = m'_h/(q_\infty c(y)^2)$ on the *local* section chord, because this wing has several defensible reference chords.
5. **Integrated:** M_h in N m and $C_{Mh,region} = M_h/[q_\infty \int c(y)^2 dy]$ over the same region, which avoids selecting one of them.
6. **Terms:** pressure is the primary result, the viscous term is reported separately, and the total is given.

Result. Table 18 gives every geometry at every condition that carries a solved trim. **Every candidate raises the hinge moment above the rigid baseline, at all three operating points, without exception.** The increase is 2.0 to 5.0 per cent at Condition CR, 0.5 to 6.3 per cent at early cruise and 0.4 to 6.2 per cent at late cruise. H is the largest at every condition, which is what a discrete hinged surface should do: it turns a rigid segment against the flow rather than redistributing camber smoothly. W is the smallest at both cruise points.

The ordering is preserved between the conditions, up to one near-tie. From largest magnitude to smallest it is H, M, C, F, W, B at both cruise points and H, M, C, W, F, B at Condition CR, where W and F differ by 0.0006 N m, 0.07 per cent of the baseline moment. A hinge moment measured at low speed therefore ranks the candidates for cruise, as the drag does (Section 5.2.3).

Both orderings rank the candidates on the common comparison line, not on the load each concept’s own actuator carries. For C, F and M those are the same thing; for H and W they are not, and W’s differs in region as well as in axis. Section 5.6.1 states which rows are actuator moments and which are comparisons only. Read it before using any row to size a mechanism or to rank the concepts against one another on actuator load.

The viscous term is small and it is reported, not assumed. It runs 0.37 to 0.44 per cent of the total at Condition CR and 0.13 to 0.17 per cent at cruise. It is computed from the sampled wall-shear vector, whose sign convention is measured rather than inferred: OpenFOAM’s `wallShearStress` is the negative of the traction on the body, and a wall shear that integrates to a thrust along the freestream is refused outright.

Reading the two figures. Both carry one panel per condition, and in both the upper row is *dimensional* while the lower row is the coefficient. The three conditions sit at $q = 708.05, 11,292.80$ and $8,257.37$ Pa, so an M_h or an m'_h from one panel may not be read against another; $C_{Mh,region}$ and c_{mh} may, which is why both rows are drawn. Sectional loading falls monotonically outboard in every case, following the chord, and the candidates separate from the baseline over the commanded span.

The sectional lines are smoothed for display and the integral is not smoothed at all. The curves carry a Hann average over 8 per cent of the band. What it removes is an artefact of the reporting bins rather than a feature of the flow: the raw values alternate from station to station at a lag-one autocorrelation of -0.49 on all three baselines, and a block rebin to 91 stations left that at -0.49 while shrinking only the amplitude, which is what established it is not a physical oscillation. Smoothing moves it to $+0.89$ and takes the direction reversals from 427 to 1, leaving the distribution’s shape untouched. The unsmoothed stations are kept in the per-case records under `results/derived/`. Nothing is smoothed before integration: M_h in Table 18 is a direct sum over faces and never passes through these stations.

5.6.1 Which of these numbers are actuator moments, and which are only comparisons

All six geometries are integrated about 0.62 because the agreed definition fixes one common comparison line, and $C_{Mh,region}$ means nothing unless all six share it. That makes every number in Table 18 a valid *comparison* on a common basis. It does not make all of them actuator moments, and the distinction is not cosmetic for anyone sizing a mechanism.

Three of the five morphed candidates act on the line integrated about. C, F and M are continuous trailing-edge camber: the morph starts at $x/c = 0.62$, the surface aft of that line is the surface that moves, and the moment reported is the moment their actuator reacts.

Table 18: Hinge moment at trim, every geometry and condition, all integrated about the common comparison line $x_h/c = 0.62$ over $0.60 \leq \eta \leq 1.00$. Δ is against that condition's own baseline B. **The last column is load-bearing: only the rows marked “yes” are the moment that concept’s own actuator reacts.** (a) H’s hinge is at $x/c = 0.70$, so the strip from 0.62 to 0.70 lies forward of it and is inside this integral but outside H’s actuator load. (b) W rotates the whole section about $x/c = 0.25$, so its actuator reacts a *torsional* moment over the entire section rather than a hinge moment about 0.62 over the aft 38 per cent; that quantity is not computed in this campaign. Rows (a) and (b) are valid comparisons on a shared line and must not be used to size a mechanism or to rank concepts on actuator load. Source: `fig/hinge_moment_values.csv`, rows generated by `scripts/make_cruise_tables.py`.

Geometry	M_h [N m]	ΔM_h	Δ %	$C_{Mh,region}$	visc. %	actuator?
<i>Condition CR, $M = 0.10$, $C_L = 0.428277635108$, $q = 708.05$ Pa</i>						
B baseline EET AR12	-0.8668	+0.0000	+0.0	-0.035973	0.41	rigid
C cte_i002_c04	-0.9022	-0.0354	-4.1	-0.037457	0.39	yes
F cfft_b02_c01	-0.8842	-0.0174	-2.0	-0.036682	0.40	yes
H chc_g02_c06	-0.9102	-0.0433	-5.0	-0.037756	0.37	no ^a
M mcv2_i002_c01	-0.9047	-0.0379	-4.4	-0.037560	0.39	yes
W cffw_b01_c01	-0.8848	-0.0179	-2.1	-0.036728	0.44	no ^b
<i>Early cruise, $M = 0.78$, $C_L = 0.529297087450$, $q = 11,292.80$ Pa</i>						
B baseline EET AR12	-16.8188	+0.0000	+0.0	-0.043763	0.15	rigid
C cte_i002_c04	-17.6537	-0.8348	-5.0	-0.045952	0.13	yes
F cfft_b02_c01	-17.1109	-0.2920	-1.7	-0.044531	0.14	yes
H chc_g02_c06	-17.8711	-1.0523	-6.3	-0.046505	0.13	no ^a
M mcv2_i002_c01	-17.7165	-0.8977	-5.3	-0.046115	0.13	yes
W cffw_b01_c01	-16.9087	-0.0898	-0.5	-0.044030	0.16	no ^b
<i>Late cruise, $M = 0.78$, $C_L = 0.544114800210$, $q = 8,257.37$ Pa</i>						
B baseline EET AR12	-12.0834	+0.0000	+0.0	-0.042999	0.15	rigid
C cte_i002_c04	-12.6916	-0.6082	-5.0	-0.045180	0.14	yes
F cfft_b02_c01	-12.2892	-0.2059	-1.7	-0.043740	0.15	yes
H chc_g02_c06	-12.8307	-0.7473	-6.2	-0.045662	0.14	no ^a
M mcv2_i002_c01	-12.7367	-0.6533	-5.4	-0.045340	0.14	yes
W cffw_b01_c01	-12.1354	-0.0520	-0.4	-0.043217	0.17	no ^b

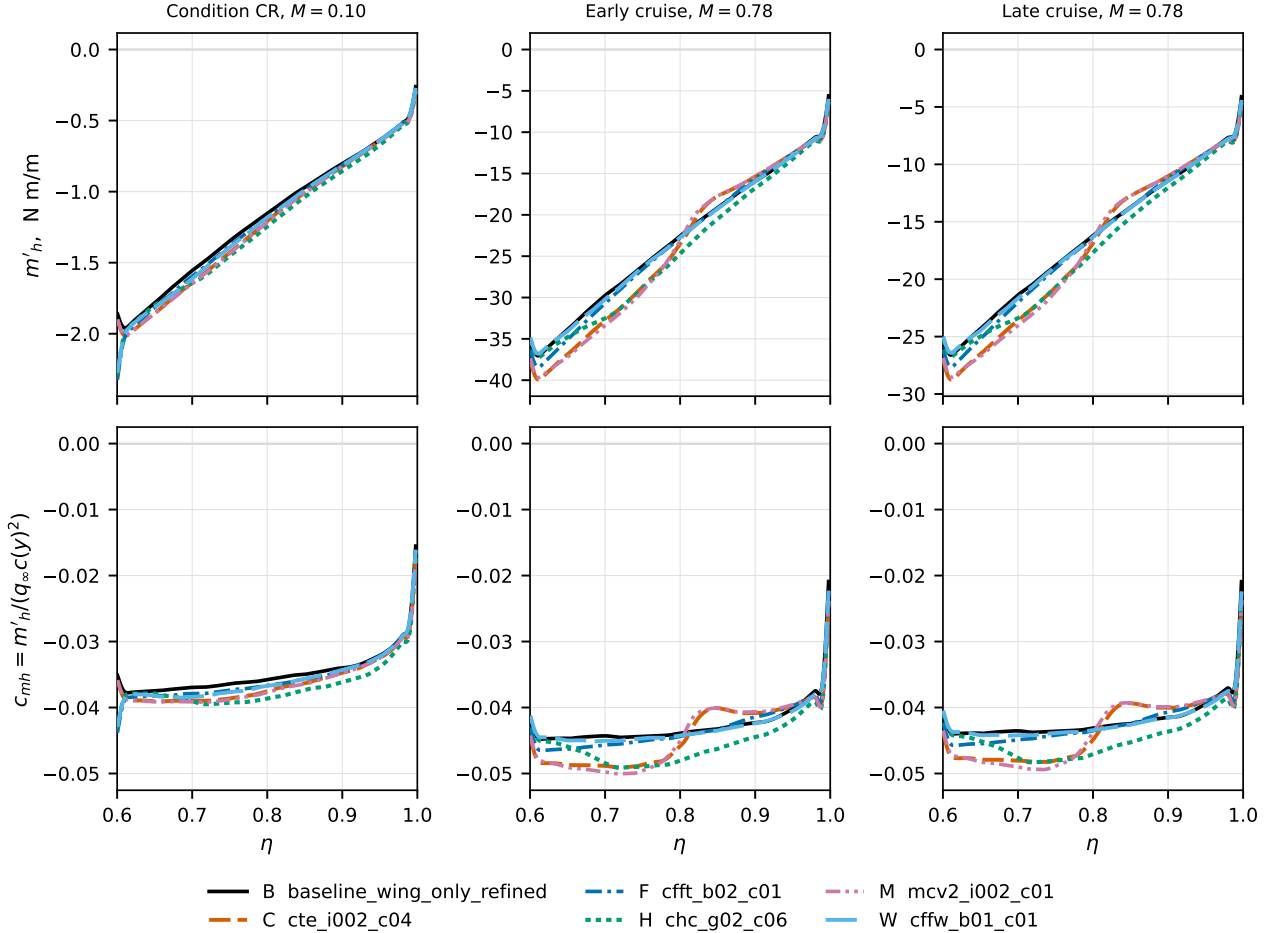


Figure 11: Sectional hinge moment against η . Upper row m'_h , lower row c_{mh} on the local chord. Lines are smoothed for display.

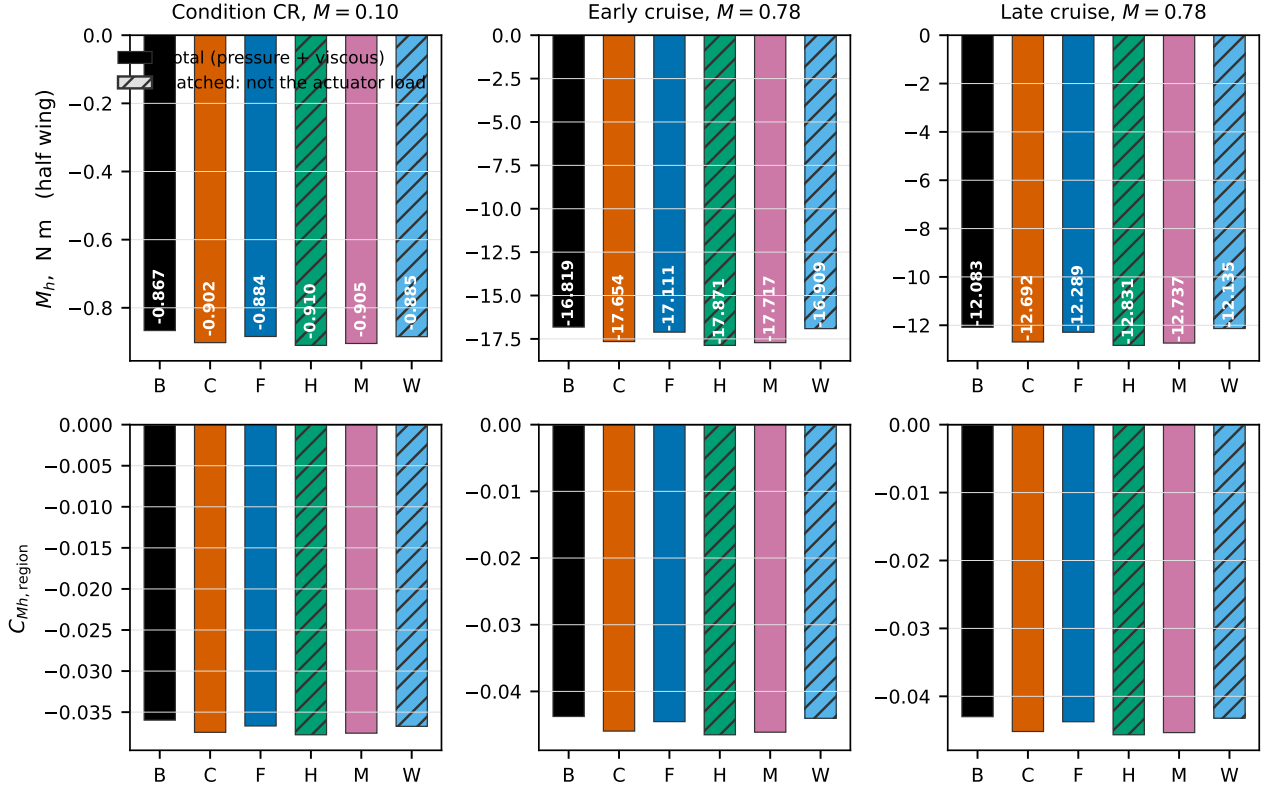


Figure 12: Integrated hinge moment, half wing. Upper row M_h , lower row $C_{Mh,region}$. The printed value is the total; the viscous share is annotated.

H's own hinge is not at $x_h/c = 0.62$. It is at 0.70 (Section 2.4). The strip between $x/c = 0.62$ and 0.70 lies *forward* of its real hinge, so it is inside the integral reported here and outside the load H's own actuator carries. H's number is therefore correct as a comparison and is not H's actuator moment.

W is not a hinged concept at all, and its row is the one most easily misread. Distributed twist rotates the whole local section about $x/c = 0.25$, and it is the one candidate whose leading edge moves (Section 2.4). Its actuator reacts a *torsional* moment, taken about the quarter chord and over the *entire* section, whereas the quantity tabulated here is taken about 0.62 and over the aft 38 per cent only. The axis is displaced by $0.37c$ rather than H's $0.08c$, and the region of integration is wrong as well as the line. **W's number is a common-basis comparison and is not, even approximately, W's actuator moment.** The torsional moment about $x/c = 0.25$ has not been computed in this campaign; it is the same integral over a different region about a different line, and it is the number the structures and actuator group needs before distributed twist can be sized or ranked against trailing-edge morphing on actuator load.

5.6.2 Method uncertainty, measured

The spanwise resolution is not free, and it was quantified rather than chosen. The hinge line is built from a planform binned in y , and the local chord is taken as the extreme x within each bin, so a coarse bin overstates the chord, moves $x_h = x_{LE} + 0.62c$ against the surface, and changes which faces count as aft of the line. Sweeping the bin count on `SWB.trim` gives $M_h = 0.9375, 0.8972, 0.8774, 0.8677$ and 0.8630 N m at 100, 200, 400, 800 and 1600 bins, with the selected face count falling from 523,627 to 501,559 and the mean chord from 0.2217 to 0.2135 m while the mean x_h holds to 0.015 per cent. The increments halve on each doubling, so the convergence is first order and Richardson puts the limit near 0.858 N m.

Every number in this section is at 1600 bins, which sits about 0.5 per cent from that limit, and 0.5 per cent is the figure to quote as the method uncertainty on an absolute M_h . The module's default of 200 bins was 4.5 per cent high in magnitude and is superseded. The *orderings* are unaffected: both condition orderings are identical at 200 and at 1600 bins, so the ranking was robust to a bias the magnitudes were not.

Late cruise. All eighteen condition-by-geometry cells are integrated, so the third operating point carries its full six-way comparison. Every candidate raises the magnitude of the moment about the common line, by 0.4 per cent on W to 6.2 per cent on H, the same direction and nearly the same size as at early cruise, and the ordering is identical at the two cruise points. The actuator-load comparison between these concepts therefore does not depend on which cruise condition is chosen.

5.7 Root bending, against the low-order method and against its screen

Purpose. Root bending is one of the two inequality screens the design study selected on, and the study’s own report calls it *the active screen* for the low-speed selection. Unlike the hinge moment of Section 5.6, the low-order method *does* produce it, so this is the one structural quantity where the two methods can be compared directly. This section asks two things: do they agree, and does the viscous answer still satisfy the screen.

What is compared, and on which basis. The RANS value is the half-wing moment about the root chord line, taken from each case’s own solver moment output about a centre of rotation asserted to lie on the symmetry plane. It is not integrated from a surface: root bending needs no chordwise resolution, so the solver’s own M_x is both the more direct route and the one available for every case. The VLM value is the half-wing root bending in each delivered optimisation record.

Two normalisations have to be stated or the comparison is meaningless. First, **root bending is dimensional**. The VLM decks run at 100 ft/s and this campaign at 34 m/s, which is $q = 569.03$ against 708.05 Pa, a ratio of 1.24431. Comparing the two moments without that factor is wrong by 24 per cent before any physics enters. Second, **the two percentages are against different baselines**: the VLM differences are against the corrected-interpolation baseline B*, whose own record states it sits -1.67 per cent from the original, while the RANS differences are against B, the surface that was actually meshed and solved. The percentage is the primary comparison because it cancels q ; the dimensional column is reported beside it precisely because a baseline disagreement would otherwise hide inside the percentages.

Result: the two methods agree to better than half a percentage point. Table 19 gives both. Every candidate agrees within 0.40 percentage points, and after the q ratio the dimensional moments agree to between 0.12 and 0.88 per cent. That is far closer than the drag comparison of Section 6.2, and the reason is physical: root bending is the first moment of the spanwise lift distribution, which a vortex-lattice method resolves well, and viscosity contributes very little to it. The lift centroid moves outboard with the morphing exactly as the moment does, from $\eta = 0.3985$ on the baseline to 0.4286 on M.

Result: the viscous calculation breaks the screen on C. The screen is a 6.8 per cent increase over the rigid baseline. C is the design study’s strict optimum and sits at +6.796 per cent in the VLM, which is 99.94 per cent of the screen: the study’s own report says the optimum “lies immediately below” it. **The RANS puts C at +7.170 per cent, which is over.** M was already over in the low-order method, at +7.124, and the RANS agrees and puts it further over at +7.519. F, H and W are comfortably inside on both methods.

What that does and does not mean. The design study states the 6.8 per cent figure is “a study-level bound rather than a certified structural limit”, and that its root bending is approximated. So this is a flag for the structures and actuator group, not a failure against a certified allowable. What it does establish is that the constraint which *selected* C is not satisfied once the same quantity is computed viscously, and that the margin it was selected on, 0.004 percentage points, was far smaller than the 0.37 percentage point difference between the two methods.

Table 19: Half-wing root bending at Condition CR. VLM percentages are against B*, RANS against B. “VLM×ratio” scales the VLM moment by $q_{\text{RANS}}/q_{\text{VLM}} = 1.24431$. Source: `fig/rootbending_values.csv`.

	Geometry	VLM %	RANS %	diff (pp)	VLM×ratio	RANS [Nm]	dim. %
B	baseline	+0.000	+0.000	+0.00	137.805	137.133	+0.49
C	cte_i002_c04	+6.796	+7.170	+0.37	147.171	146.965	+0.14
F	cfft_b02_c01	+3.755	+3.960	+0.20	142.980	142.563	+0.29
H	chc_g02_c06	+3.769	+3.368	−0.40	142.999	141.752	+0.88
M	mcv2_i002_c01	+7.124	+7.519	+0.40	147.623	147.444	+0.12
W	cffw_b01_c01	+3.010	+3.079	+0.07	141.953	141.355	+0.42

5.7.1 One delivered percentage is stale, and it is the one that matters

M’s delivered increase is wrong in the file. mcv2_i002_c01 records a moment of 118.64405 Nm and an increase of 5.3352 per cent. Those two are inconsistent: $118.64405/110.75354 = 1.07124$, an increase of 7.124 per cent. The moment is the corrected value while the percentage was formed against the pre-correction baseline. The design study’s own report states the correct figure in prose, “the unchanged candidate now increases half-wing root bending by 7.12 per cent, above the illustrative 6.8 per cent screen”, so this is a known defect in the delivered record rather than a new disagreement. **Every percentage in Table 19 is recomputed from the two moments rather than read**, and the generator carries the delivered value alongside and flags the disagreement, so the discrepancy is visible rather than silently corrected. A reader taking the number straight from the CSV would conclude M passes the screen by 1.5 percentage points when it fails it by 0.3.

5.7.2 Cruise, and why there is no low-order column

The five candidates do not appear in the study’s cruise set at all. That set contains optimiser seeds and the rigid baseline, indexed by concept rather than by the geometries meshed here, so there is no VLM root bending to compare against at either cruise point and none is constructed. The cruise numbers below are RANS alone.

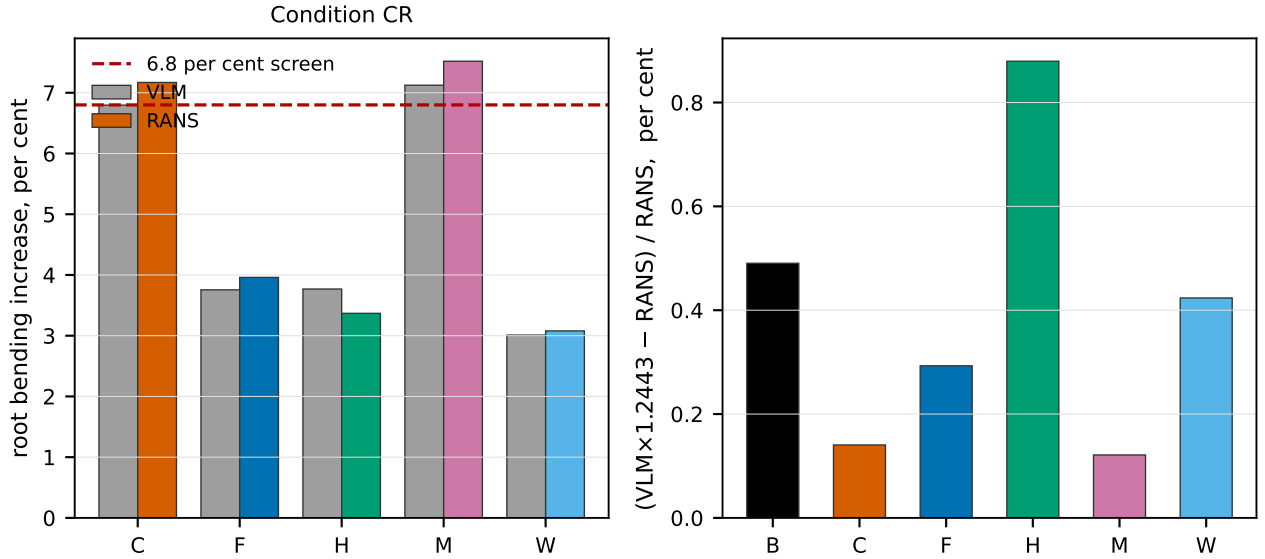


Figure 13: Root bending increase against each method’s own baseline, with the 6.8 per cent screen; and the dimensional agreement after the q ratio.

Applying the study’s own definition of the cruise limit, which is the cruise rigid baseline’s own moment at that operating point, so that utilisation is the ratio to it:

	B	C	F	H	M	W
Early cruise M_x [N m]	2632.22	2803.09	2748.73	2727.70	2803.51	2720.56
Early cruise utilisation	1.000	1.065	1.044	1.036	1.065	1.034
Late cruise M_x [N m]	1978.59	2102.15	2063.14	2047.23	2101.72	2042.97
Late cruise utilisation	1.000	1.062	1.043	1.035	1.062	1.033

The limit is not the 6.8 per cent screen, and carrying that number forward from Section 5.7 inverts every verdict in this table. The screen applied at Condition CR admits a 6.8 per cent increase over the rigid baseline. The cruise limit is the study’s own and is stricter: it is the baseline moment itself, so the admissible increase is 0 per cent and any utilisation above 1.000 is over. The two give opposite answers on the same numbers. On the 6.8 per cent screen every candidate here passes, C and M being the closest at +6.49 and +6.51 per cent at early cruise; on the cruise limit every candidate fails. W carries the smallest increase at both cruise points, +3.36 and +3.25 per cent. Neither bound is a certified allowable, so the question for the structures and actuator group is which to size against, and this report does not answer it.

All five exceed the cruise limit, and that is consistent with the study rather than contrary to it. These are low-speed-selected shapes evaluated at cruise, and the study reports the same effect from the other direction: transferring its late-cruise optima unchanged to early cruise takes trailing-edge camber and twist to 105.07 and 104.02 per cent of the limit. A shape selected at one operating point does not carry its bending margin to another. Late cruise gives the same ordering as early cruise, C and M highest and W lowest.

5.8 Surface pressure and skin friction

The surface distributions behind every integrated quantity above are in the appendices: the five-station cuts and their overviews in Appendix A, and the whole-wing difference maps in Appendix B. Nothing separates at Condition CR. At both cruise points the two candidates with the largest commanded camber, C and M, separate behind the recompression shock at $\eta = 0.80$ and the other four do not (Section 5.2.4). The low-order study has no counterpart to any of it.

6 Can the VLM drive the design study?

Purpose. This section measures VSPAERO against RANS on the same six geometries and answers whether the vortex-lattice method can keep driving candidate selection. The comparison is made at Condition CR, where both sides share one frame and one geometry set, and it asks two things: does the low-order method put the candidates in the order the RANS puts them, and how far off is it on the size of the effect.

6.1 Method

Method. VSPAERO is a vortex-lattice solver, and it ran the same six geometries that were meshed. Each deck is matched to its RANS case by `.vsp3` SHA-256, never by name. All nine Condition CR decks read $S_{\text{ref}} = 13.3572 \text{ ft}^2$, so the conversion onto the DSO basis is exactly 1.000000. The package also carries a second baseline deck, `baseline_corrected`, which the design study ranks against; every VLM increment in this report

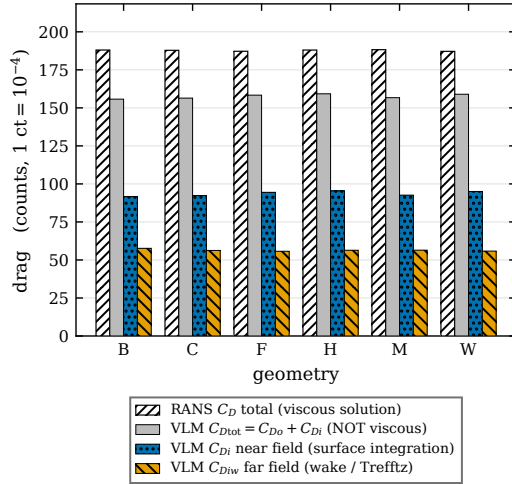
is against the deck of the meshed baseline B, and referencing it to the other deck moves all five candidates by one constant and changes no ordering.

Which columns compare. VSPAERO writes $C_{Dtot} = C_{Do} + C_{Di}$, and the decks close that identity to 10^{-12} . C_{Do} is a flat-plate skin-friction estimate, and it supplies 40.1 to 41.1 per cent of C_{Dtot} on the six meshed geometries. The decks evaluate it at $Re = 1 \times 10^7$, against $Re_{C_{ref}} = 0.92 \times 10^6$ here on the same chord. The defensible drag comparison is therefore induced against induced, and C_{Dtot} appears nowhere beside a RANS C_D . Each VLM induced increment below sits beside the RANS total increment at the same fixed C_L ; the RANS induced drag from the Trefftz planes is in Section 5.5, and its span efficiency enters Table 21 beside the VLM’s own.

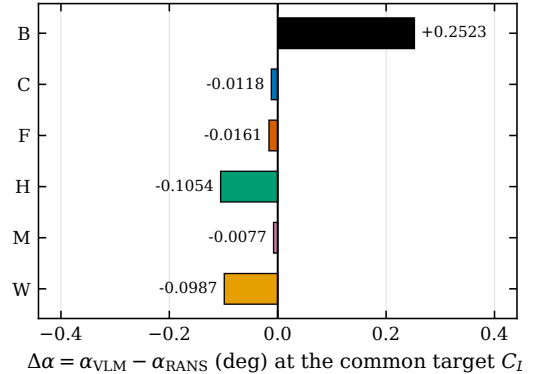
Result. Table 20 gives the VLM side, and Figure 14(a) draws its composition against the RANS total. The two induced columns differ from each other more than either differs from anything else. Near-field span efficiency runs $E = 0.57$ to 0.59 , far-field $E_w = 0.94$ to 0.97 , on the same runs. An independent lifting-line calculation on the spanwise loading returns 56.9 ct, which supports the far-field column to 0.4 per cent. Both columns are carried through the table that follows.

Table 20: VSPAERO at Condition CR, $M = 0.10$, each deck trimmed to $C_L = 0.428277635108$. Drag in counts on the DSO basis, deck $S_{ref} = 13.3572 \text{ ft}^2$, conversion onto the report basis exactly 1.000000. C_{Di} is the near-field surface integration and C_{Diw} the far-field wake integral; E and E_w are their span efficiencies. B is the surface the RANS meshes. Source: `data/vlm.json`.

Geom.	Deck	α [deg]	C_{Do}	C_{Di}	C_{Dtot}	C_{Diw}	E	E_w
B	baseline_wing_only_refined	1.99981	64.060	91.676	155.736	57.649	0.5907	0.9394
C	cte_i002_c04	1.25652	64.078	92.342	156.421	56.258	0.5865	0.9627
M	mcv2_i002_c01	1.24972	64.084	92.617	156.701	56.441	0.5847	0.9595
F	cfft_b02_c01	1.45936	63.894	94.465	158.359	55.712	0.5733	0.9721
W	cffw_b01_c01	1.42193	63.927	95.030	158.957	55.827	0.5699	0.9701
H	chc_g02_c06	1.44697	63.887	95.370	159.257	56.383	0.5679	0.9605



(a) Drag decomposition.



(b) Trim angle.

Figure 14: Condition CR, $M = 0.10$, $C_L = 0.428277635108$, DSO basis, RANS half model $A_{ref} = 0.620462 \text{ m}^2$. (a) What each method contains: the RANS total beside the VSPAERO $C_{Dtot} = C_{Do} + C_{Di}$ split into its parasite correlation and its induced part, with the far-field C_{Diw} alongside. The two bar heights are not differenced, because C_{Do} is a flat-plate correlation at the deck Reynolds number. (b) Trim angle at the common target C_L , which is the closest agreement in this section.

6.2 Drag increments

Result. Table 21 is the central comparison of this section. The RANS morphing signal spans -0.868 to $+0.232$ ct, 1.100 in total, and neither VLM column resolves an effect that small. The near-field column misses the measured increment by 0.709 to 4.222 counts and gets the sign wrong on all four candidates the RANS puts below the baseline. The far-field column misses by 0.955 to 1.441 counts, every error of the same sign and within half a count of the others, which is why its ordering survives where the near-field ordering does not: a near-constant offset moves the whole field together and leaves the sequence intact.

Far field. On the far-field column every predicted increment is negative. The VLM has all five candidates saving induced drag, while the RANS measures four of the five costing total drag. Those errors run 1.24 to 7.85 ct, and for F, W and H they exceed the entire signal span. Part of that gap is a real profile increment the VLM has no mechanism to produce, and a RANS Trefftz integral is what separates the two.

Systematics. The RANS-side systematic uncertainty is far smaller than any of these differences. Window drift is 0.0014 to 0.0028 ct against a gate of 0.05 ct, and removing each trim case’s own C_L residual moves the increments by at most 0.021 ct.

Table 21: Selection test at Condition CR, $M = 0.10$, $C_L = 0.428277635108$, DSO basis (RANS half model $A_{\text{ref}} = 0.620462 \text{ m}^2$; VSPAERO $S_{\text{ref}} = 13.3572 \text{ ft}^2$, conversion exactly 1.000000). Increments are against the baseline B, in counts: the RANS increment is a total drag increment and the VLM increments are induced drag increments, so “error” is the VLM prediction minus the RANS measurement. Ranks are by increasing increment against each method’s own baseline, rank 1 best; the closest adjacent RANS pair is W and F at 0.053 ct, so no pair is tied. The last three columns are span efficiencies: the VLM’s own near- and far-field values, and the RANS value at the last Trefftz station. Rows and statistics generated by `scripts/make_rank_table.py` from `fig/fig_vlmerr_values.json`, `data/vlm.json` and `data/span_efficiency.json`.

Geom.	RANS		VLM near field			VLM far field			span efficiency		
	ΔC_D	rank	ΔC_{Di}	rank	error	ΔC_{Diw}	rank	error	VLM E	VLM E_w	RANS e
W	−0.868	1	+3.354	4	+4.222	−1.823	2	−0.955	0.5699	0.9701	0.9837
F	−0.815	2	+2.789	3	+3.604	−1.938	1	−1.123	0.5733	0.9721	0.9859
C	−0.191	3	+0.666	1	+0.857	−1.392	3	−1.201	0.5865	0.9627	0.9773
H	−0.009	4	+3.694	5	+3.703	−1.266	4	−1.257	0.5679	0.9605	0.9773
M	+0.232	5	+0.941	2	+0.709	−1.209	5	−1.441	0.5847	0.9595	0.9724
Kendall τ , all five			−0.20			+0.80					
Spearman ρ , all five			−0.20			+0.90					
Pairs swapped, of 10			6			1					

6.3 Is the ranking preserved?

Result. Ranking is the property a search tool needs, and it survives on one column (Table 21), but not the column the June selection was made on. The RANS orders the candidates $W < F < C < H < M$. Far-field C_{Diw} gives $F < W < C < H < M$, nine of ten pairs concordant, Kendall $\tau = +0.80$. Near-field C_{Di} gives $C < M < F < W < H$, four of ten concordant, $\tau = -0.20$.

Which way the two metrics fail. The far-field metric’s single error is on the closest pair in the field: the RANS separates W and F by 0.053 counts, the narrowest adjacent gap of the five, and against a convergence gate of 0.05 counts per window that pair is at the edge of what this campaign can resolve. A metric that orders everything except the one pair nobody can separate is performing about as well as the data permits. The near-field metric fails differently and not by a small margin: $\tau = -0.20$ is worse than no information, because it groups $\{C, M\}$ as the cheap pair and $\{F, W, H\}$ as the expensive one, and the viscous answer is close to the reverse.

What this claim is not. The Condition CR field spans 1.100 counts and the convergence gate is 0.05 counts per window, so the spread is twenty-two gates wide but the closest adjacent pair is one. Reproducing this ordering is therefore evidence that a metric tracks the sign and the grouping of the field, not that it resolves neighbours within it; Section 5.1 makes the same point from the other side, that the low-speed sequence is read as a group. The two statements are consistent and the distinction is the whole of it: a search tool that puts the cheap candidates in the cheap group is useful even where it cannot separate two of them, and that is what $\tau = +0.80$ with its one error on the narrowest pair describes. **Reading.** The two induced-drag columns the VLM writes are not interchangeable, and this is where the choice between them shows. The far-field wake integral is the one that tracks the viscous ordering; the near-field surface integration does not.

6.4 Decision

Decision. The VLM can be trusted to order candidates, on its far-field column only, and to price none of them. Ordering: far-field C_{Diw} reproduces the RANS order to nine pairs of ten, near-field C_{Di} to four of ten. Pricing: the whole morphing effect at Condition CR spans 1.100 counts, from W at −0.868 to M at +0.232, and the far-field increments miss the measured ones by 0.955 to 1.441 counts, which is 87 to 131 per cent of that span. The near-field increments miss by 0.709 to 4.222 counts, 64 to 384 per cent.

Why one of them still ranks. The five far-field errors are all of one sign and lie within half a count of each other, −0.955 to −1.449. An error that is nearly a constant offset moves every candidate together and therefore leaves the ordering intact while destroying the magnitude, which is precisely the behaviour seen. The near-field errors are neither one-signed in magnitude nor clustered, spanning a factor of six, and the ordering does not survive them.

Consequence. Candidate search can stay with VSPAERO provided it selects on the far-field column, which is the column the design study adopted in its August correction and not the one the June selection was made on. Every count quoted for a selected concept comes from RANS, and at Condition CR that is not a refinement but a requirement: the low-order error is the size of the entire effect being resolved.

Consequence for selection. Both metrics predict a saving for every candidate at Condition CR and the viscous calculation returns a saving for four of the five. At cruise it returns a saving for W alone and a cost for the other four (Section 5.2). **The three geometries the low-order study currently selects, W, F and H, are the three cheapest at both cruise points, in that order.** C and M, from the July selection on the

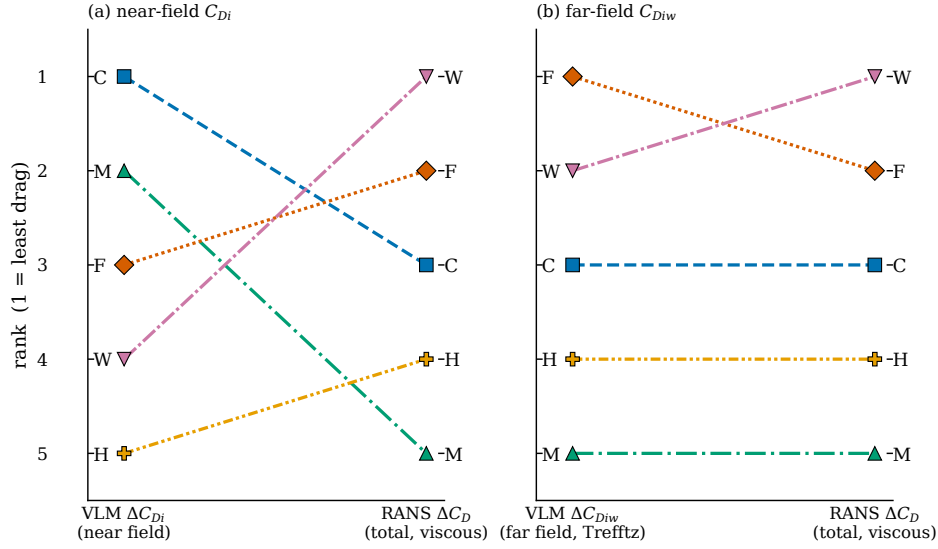


Figure 15: Selection test at Condition CR, $M = 0.10$, every geometry trimmed to $C_L = 0.428277635108$ on the DSO basis. Each panel ranks the five candidates by drag increment, low order on the left and RANS on the right, rank 1 being the least drag. Panel (a) uses near-field C_{Di} , the metric the June selection was made on, and swaps 6 of 10 pairs. Panel (b) uses far-field C_{Diw} , the metric the design study now selects on, and swaps 1 of 10, that pair being W against F at 0.053 counts apart. Generated by `fig/make_vlmerr.py` from `data/vlm.json` and the Condition CR RANS trims.

near-field column, cost twelve to fifteen counts. The cruise cost of C and M is wave drag and separation behind a strong aft shock (Section 5.2.4), which no induced-drag column can see, so the selection is right at cruise for a reason the low-order method does not model. Reselection that admits strong-camber shapes should therefore rank on a metric that sees the shock; of the two low-order induced-drag columns, the far-field one is the only one that orders the candidates at low speed.

7 What this report establishes, and what it does not

Purpose. This section states the whole result in one place, after the loads that bear on it, so that no single section can be read as the conclusion on its own.

7.1 What is established

- Drag: W is the cheapest geometry at every operating point, and the low-speed ordering carries to cruise.** At Condition CR four candidates improve on the rigid baseline, W by 0.868, F by 0.815, C by 0.191 and H by 0.009 counts against 188.031, and the whole field spans 1.1 counts, enough to separate the groups but not the neighbours within them. At early cruise W saves 0.769 counts against 177.839, F and H cost 1.796 and 2.244, and C and M cost 12.871 and 14.554; late cruise reproduces it, W at -0.781 to M at $+13.728$ against 187.959. **The two cruise points agree exactly**, both giving W, F, H, C, M, and they differ from the low-speed sequence W, F, C, H, M only in the C and H pair, which is 0.182 counts apart at Condition CR (Section 5.2.3).
- The cruise cost is the aft shock.** C and M end the supersonic region at $\eta = 0.80$ in a strong shock and separate behind it; F and H in a moderate shock and stay attached; W and the baseline without a shock. Drag follows that order exactly (Section 5.2.4).
- Hinge moment: every candidate raises the actuator load, and two of the rows are not actuator loads at all.** All five raise the moment about the common $x_h/c = 0.62$ line at all three operating points, by 2.0 to 5.0 per cent at Condition CR, 0.5 to 6.3 at early cruise and 0.4 to 6.2 at late cruise, with the same ordering at all three up to one near-tie. C, F and M act on the line integrated about, so their numbers are the moment their actuator reacts. **H's and W's are not:** H's hinge is at 0.70, and W is not hinged at all but rotates the whole section about $0.25c$ (Section 5.6.1).
- Root bending: the constraint that selected C is not satisfied viscously.** C sits at $+6.796$ per cent in the VLM, inside the study's 6.8 per cent screen by 0.004 percentage points, and the RANS puts it at $+7.170$ per cent, outside. M is outside on both. F, H and W are comfortably inside on both. At both cruise points, under the study's own stricter limit, all five exceed the baseline moment, W by the least (Section 5.7.2).
- The two methods agree on root bending to better than half a point.** VLM and RANS differ by at most 0.40 percentage points, so the screen result above is a property of the shapes rather than of the method.
- The low-order method orders the candidates on its far-field column and prices none of them.** Far-field C_{Diw} reproduces the RANS order at Condition CR to nine pairs of ten, its one error on the closest

pair in the field; near-field C_{Di} , the metric the June selection was made on, gives four of ten. Neither resolves the size of the effect: the far-field increments miss the measured ones by 86 to 131 per cent of the whole low-speed span and the near-field by 63 to 381 (Section 6.4).

7.2 What is not established, stated as plainly

1. **W's actuator load is not computed anywhere in this report.** Distributed twist reacts a torsional moment about $x/c = 0.25$ taken over the entire section. The quantity tabulated for W is about 0.62 over the aft 38 per cent, which is a valid comparison on a shared line and is not that load. Until the torsional moment is computed, **trailing-edge morphing and distributed twist cannot be ranked against one another on actuator load**, which is the axis the concept selection was expected to turn on.
2. **H's actuator load is likewise not its tabulated number**, for the smaller reason that its own hinge is 0.08c aft of the comparison line.
3. **Neither root-bending bound is a certified allowable.** The design study calls the 6.8 per cent figure "a study-level bound rather than a certified structural limit". The cruise limit is a different and stricter definition. Which to size against is a question for the structures and actuator group and is not answered here.
4. **The low-order comparison is made at Condition CR only.** The VSPAERO decks and the meshed geometries share one frame and one geometry set there and nowhere else: the cruise decks carry the TP-1580 S_{ref} and a geometry set that maps to none of the meshed cases, and the study's cruise set contains optimiser seeds and the rigid baseline rather than the five candidates. Extending the comparison to either cruise point requires the corresponding VLM columns at that operating point, not merely the RANS ones, and quoting the two together without them would cross operating points (D076).
5. The meshed baseline is the uncorrected aerofoil-interpolation lineage and all five candidates are the corrected one. Measured directly between the two .vsp3 files by `scripts/measure_lineage_pair.py` into `data/lineage_measured.json`, the uncorrected wing carries 9 distinct aerofoils across its 19 stations where the corrected wing carries 19, so the two surfaces agree exactly at every station they share and diverge only in between. They are identical at and inboard of $\eta = 0.450$; the first station that differs is $\eta = 0.515$, which is also the worst, at 0.407 percentage points of t/c and 0.209 per cent of chord in surface ordinate. The camber difference is an order of magnitude smaller again, at most 0.0034 per cent of chord. Both wings are the same planform at the same twist, so this is a thickness-channel difference, and the candidate-to-candidate comparison is taken between surfaces of the corrected lineage throughout and does not cross it.
6. Condition CR runs kOmegaSST wall resolved and both cruise points run Spalart–Allmaras wall modelled, so counts are differenced within a condition only.
7. Every case is fully turbulent from the leading edge and in free air, so a tunnel comparison needs blockage, wall-interference and support corrections on the measured side.
8. The VSPAERO columns are induced-drag increments and the RANS columns are total-drag increments, so Table 21 compares ordering rather than magnitude.
9. The cruise drag-due-to-lift slopes are formed on the trim legs alone, and six of the twelve rest on a nudge too small to resolve the slope. They are used only to convert trim residuals of at most 0.895 counts of C_L into a drag bias of at most 0.036 counts.

7.3 Recommendation

Recommendation. W is the geometry to carry forward. It is the cheapest of the six at all three operating points, 0.868, 0.769 and 0.781 counts below the baseline at Condition CR, early cruise and late cruise, it raises L/D at both cruise points, and it carries no aft shock and no separation at cruise. It also has the smallest root-bending increase of the five at both cruise points, +3.36 and +3.25 per cent, and sits inside the 6.8 per cent screen at Condition CR on both methods. F and H are the next two, within 2.3 counts of the baseline at cruise. C and M are not candidates for cruise: they cost twelve to fifteen counts, separate behind a strong aft shock, and both fail the root-bending screen once it is computed viscously.

The caveat on W, and it is the one this report cannot close. W's actuator reacts a torsional moment about the quarter chord over the whole section, and that moment is not computed here (Section 5.6.1); its tabulated hinge moment is a common-basis comparison only. W's aerodynamic case is complete. Its mechanical case is not, and it is the input the structures and actuator group needs before W can be committed to.

7.4 The one number that would change the picture

W's torsional moment about $x/c = 0.25$ over the full section, at every operating point. It is the same integral this report already performs, over a different region about a different line, and it is the missing input to the concept selection, and the recommendation above rests on it. Nothing else outstanding has comparable leverage: the drag case is closed at all three operating points, and the root-bending case is closed against both bounds.

A Surface pressure and skin friction, by section

Purpose. This appendix carries the surface distributions behind the integrated coefficients in the body of the report. Every geometry is cut at five spanwise stations, $\eta = 0.20, 0.40, 0.60, 0.80$ and 0.95 , at its own trimmed incidence to that condition's own target lift: $C_L = 0.428277635108$ at Condition CR, $C_L = 0.529297087450$ at early cruise and $C_L = 0.544114800210$ at late cruise, all on the DSO basis. Comparing the six geometries' $-c_p$ peaks at a station means something only because they share that target within a condition.

Conditions are never overlaid. Condition CR and cruise are normalised on different dynamic pressures: $q = 578$ (kinematic, m^2/s^2 , the incompressible solver carries no density) against $q = 11,292.8$ Pa at early cruise and $8,257.37$ Pa at late cruise. Each is correct and they are not comparable curves, so every condition is plotted as its own set and the plotting script refuses to mix them.

Coverage. All three conditions are six geometries of six: 90 station files over 18 cases, with no case partially represented.

What is plotted. Figures 16, 17 and 18 carry c_p on an inverted axis and its difference from the baseline by surface; Figures 19 and 20 carry the signed skin friction by surface. All are against x/c on the local chord of that station, which falls from 0.4816 m at $\eta = 0.20$ to 0.1671 m at $\eta = 0.95$.

The curves are a band, not a line. The pressure and skin-friction curves pool faces within ± 0.004 semispan, ± 7.32 mm, and the wing is swept and tapered, so faces at one x/c sit at different y and carry genuinely different z ; curves are the median over $0.005c$ bins per surface rather than the raw scatter. Within one bin the 10–90 percentile spread of z/c is 1.5 to 2.3 per cent of chord at the leading edge, where the surface turns through nearly 180° , and about 1.1 per cent at the blunt trailing edge, against 0.21 to 0.49 per cent at mid-chord. A defective surface would be ragged at mid-chord too.

Two of the three sources of raggedness are the sampling band. The third was a defect and is now corrected. The leading-edge scatter is the band meeting high curvature: across the five stations the within-bin spread grows by a factor of 2.79 while the band's share of the local chord grows by 2.80, with mid-chord flat throughout as a control. That is also why the outboard stations are the noisiest, the fixed 14.63 mm slab spanning 3.0 per cent of chord at $\eta = 0.20$ but 8.8 per cent at $\eta = 0.95$. The inboard trailing-edge ripple is the band meeting taper rather than curvature: the chord shortens by 0.2279 m per metre of span, so it varies by 3.33 mm across the slab and a face is binned at an x/c that is not quite its own. Neither is removable without narrowing the band, and both are sampling rather than flow. The OUTBOARD trailing-edge ripple was neither, and it was not flow either: it was the upper-lower split failing where the two surfaces converge. Splitting on the exported facet normals instead removes it. Measured as the within-bin C_p spread over $x/c = 0.92$ to 1.00 , C at $\eta = 0.80$ falls from 0.2623 to 0.0204 and M from 0.2797 to 0.0177 ; at $\eta = 0.95$ the six cases fall from about 0.34 to about 0.09 . Two controls hold and are what make this a correction rather than a smoothing: mid-chord is unchanged to four decimal places, 0.0447 against 0.0447 for C at $\eta = 0.80$, and the inboard station is unchanged at $\eta = 0.20$, ratio 1.03 , because the band and the taper are still there.

The band also over-reads the local chord, by a fixed length rather than a fixed fraction. It takes its leading edge from the inboard face of the slab and its trailing edge from the outboard one, so on a swept wing it over-reads by $(w/2)(\tan \Lambda_{LE} + \tan \Lambda_{TE})$. With $w = 14.63$ mm, $\Lambda_{LE} = 28.84^\circ$ and $dc/dy = -0.266$ that predicts 6.11 mm against 6.67 mm measured, and the measured excess is near-constant across span while the chord itself halves, which is the signature of a slab width rather than of a scaling error. The chords quoted above are the planar-cut values and are accordingly 1.4 per cent smaller than the band's at $\eta = 0.20$ and 3.8 per cent smaller at $\eta = 0.95$. Because x/c is normalised to span 0 to 1, the curves are unaffected; only the dimensional chord printed above each column of the figures differs.

The sign on $C_{f,s}$ is the useful part. $C_{f,s}$ is the component of wall shear along the freestream, taken from each case's own `dragDir`, so *negative means reversed flow*. The sign convention is calibrated per case against inboard mid-chord faces whose attachment is known independently, not assumed; it came out one-signed on all twelve cases.

A.1 Overview at the five stations

Figures 16 to 20 draw the five stations of each condition on one page each, pressure first and then skin friction by surface; the two subsections that follow read them condition by condition.

Reading the skin-friction figures. Condition CR is wall resolved at $y^+ \approx 1$ with kOmegaSST and cruise is wall modelled with Spalart-Allmaras, so c_f is not equally resolved between the two conditions and the two columns of each figure are read separately rather than against one another. Neither c_p nor c_f carries a reference area.

The plotted quantity and its sign. The figures show $c_{f,s} = m(\tau_w \cdot \hat{d})/q$, the *freestream-aligned* component of wall shear, with $\hat{d}$ taken from each case's own `forceCoeffs` drag direction. On this swept wing that is not the section-normal component. OpenFOAM returns a negative wall shear for flow attached in $+x$, so a calibrated sign multiplier m is read from every file's own header, $m = -1$ on all of them, and positive then plots as attached. It is therefore the **sign change** that locates separation, never the magnitude. Lines are medians over $0.005c$ bins of the sampled band, with at least five faces per bin. A band counts as a *separated run* only when at least

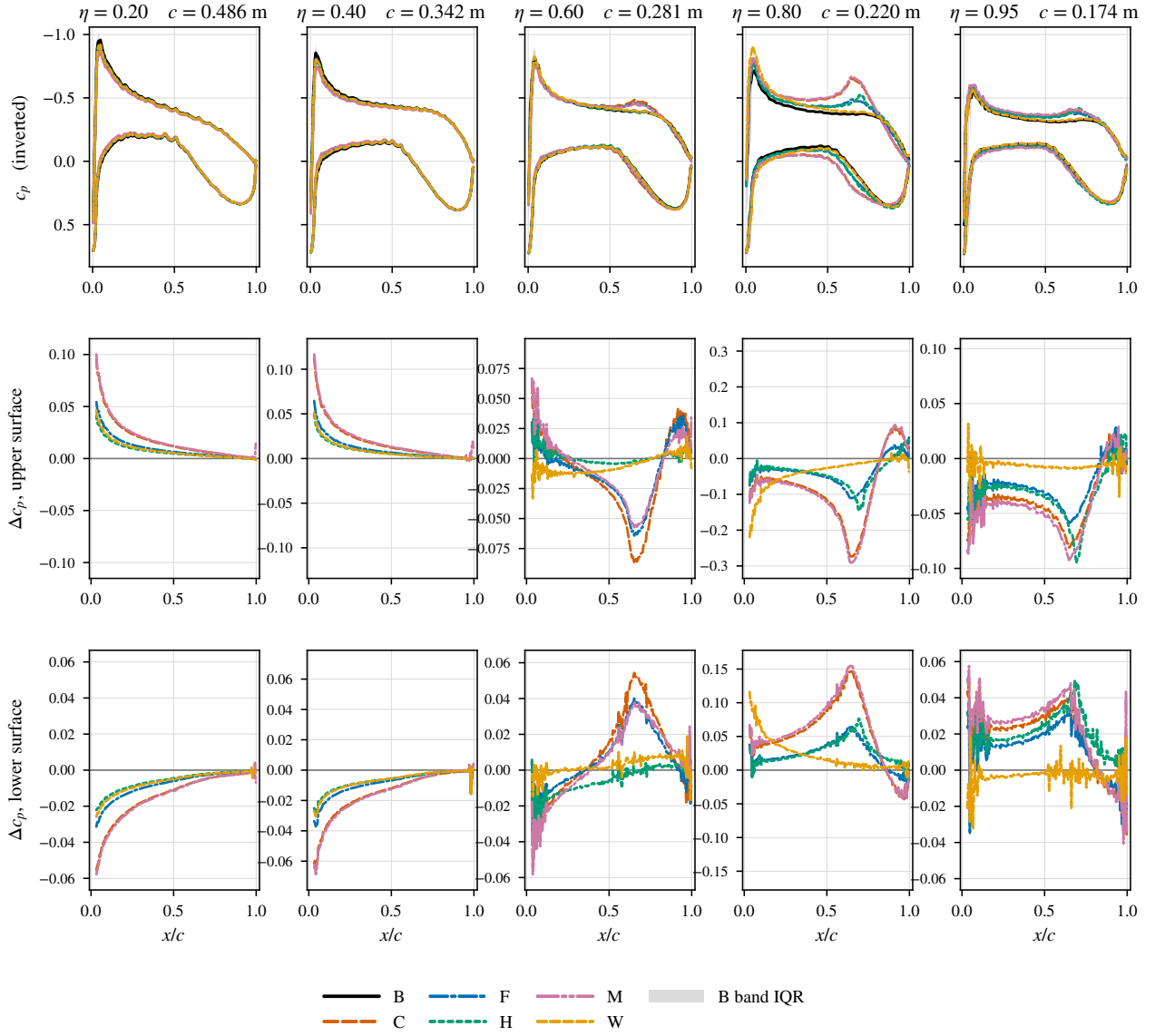


Figure 16: Surface pressure at five spanwise stations, $\eta = 0.20$ to 0.95 , Condition CR, every geometry at its own trim to $C_L = 0.428277635108$. The top row is c_p on each case's own freestream q , inverted, with the grey band the baseline's interquartile spread within the pooled span band; the two lower rows are each candidate minus the baseline on the upper and lower surfaces, drawn from $x/c = 0.03$ because forward of that the branch split and the span band are ill-conditioned. x/c is on the band's local chord, printed above each column.

four consecutive bins are negative, which is more than 2 per cent of local chord, so an isolated negative bin is not read as separation. At $\eta = 0.95$ and $x/c > 0.875$ the curve is drawn in grey because the aft bin there is band-averaged rather than resolved.

What the ninety bands contain. Over the 90 (case, station) bands of the three conditions, the largest reversed fraction on any upper surface is 25.3 per cent (LCM_trim, $\eta = 0.80$), and 31 bands carry any reversed upper-surface face at all. On the lower surface the largest is 0.474 per cent (SWC_trim, $\eta = 0.20$); 56 bands carry a trace, and those sit at the root junction and in the blunt trailing-edge base region rather than being boundary-layer separation. The four bands that genuinely separate, C and M at $\eta = 0.80$ at each cruise point, are named in Section 5.2.4.

A.2 Condition CR, $M = 0.10$, $C_L = 0.428277635108$

Nothing separates. Across all six geometries and all five stations the reversed fraction never exceeds 0.25 per cent of faces in a band, and those faces sit at the blunt trailing edge and the root junction, where they are present on the rigid baseline too. The six geometries lie almost on top of one another: at $\eta = 0.60$ the upper-surface suction plateaus agree within about 0.02 in $-C_p$, and the morphing appears only aft of the hinge.

A.3 Early cruise, $M = 0.78$, $C_L = 0.529297087450$

At $M = 0.78$ the aft-camber candidates carry a recompression shock on the upper surface between $x/c \approx 0.65$ and 0.78 that is absent from the baseline. Two of them separate behind it, and they are the two with the largest

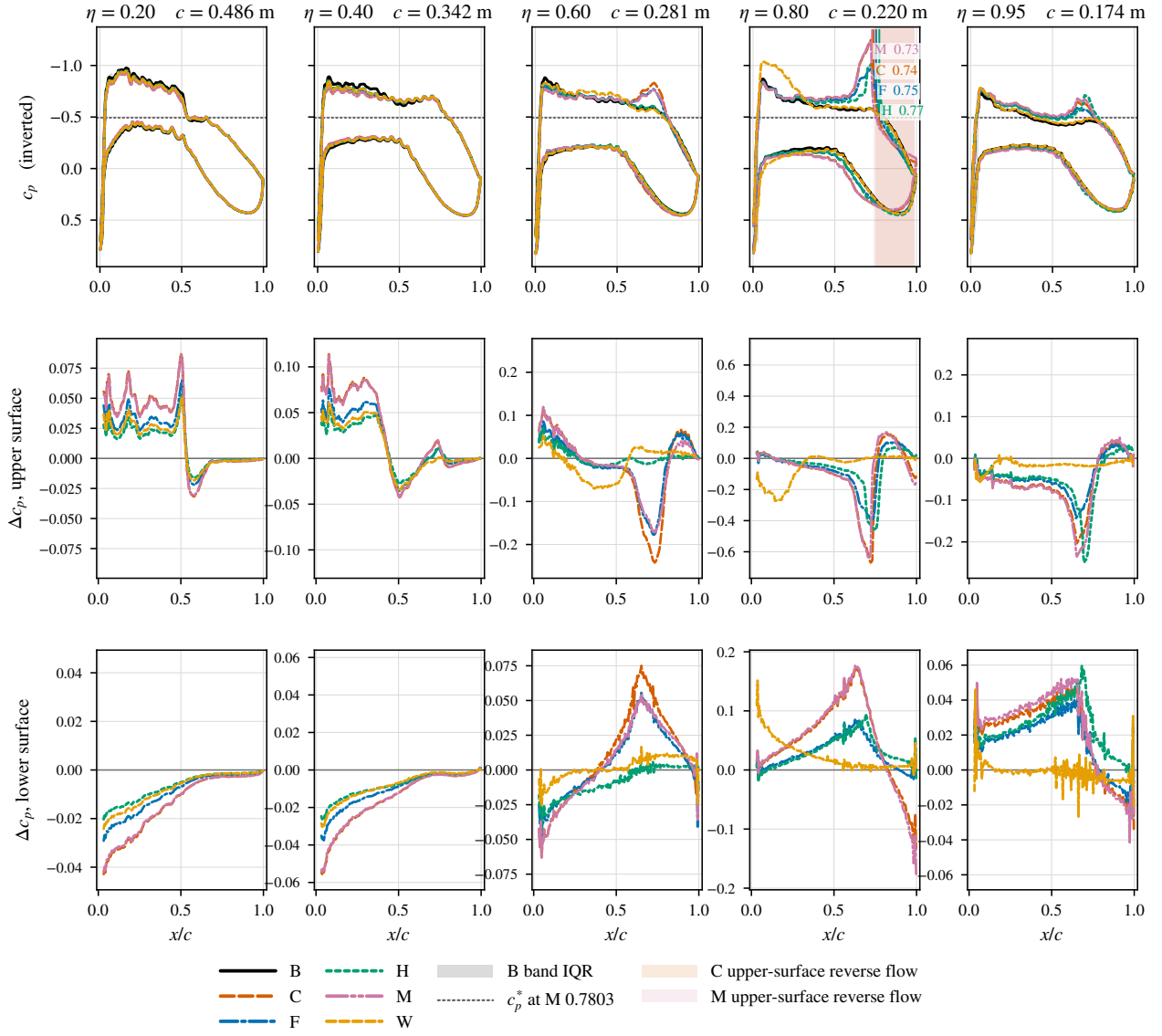


Figure 17: Surface pressure at the same five stations, early cruise, every geometry at its own trim to $C_L = 0.529297087450$; layout as Figure 16. The dashed line is the sonic c_p^* , the tick marks at $\eta = 0.80$ locate each candidate's shock by its x/c , and the shaded runs are where the upper-surface flow on C and M is reversed.

commanded camber.

At $\eta = 0.80$, C peaks at $-C_p = 1.279$ at $x/c = 0.716$ and M at 1.249 at $x/c = 0.703$. Both lose the boundary layer behind the shock and do not recover it before the trailing edge: 23.3 per cent of C's upper-surface faces in that band are reversed, over $x/c = 0.743$ to 0.992 , and 23.9 per cent of M's, over $x/c = 0.730$ to 0.997 , while both lower surfaces reverse nowhere at all. M's shock is the stronger of the two, $dc_p/d(x/c) = 50.7$ against C's 45.7, which follows its larger command: $A/c = 0.0350$ against C's 0.0337. F and H carry a weaker shock at a lower peak, 0.988 at $x/c = 0.721$ and 1.027 at $x/c = 0.734$, and stay attached with zero reversed faces; F's command is $A/c = 0.0139$, two and a half times smaller. W peaks forward, 1.074 at $x/c = 0.067$, and has no aft shock at all. The baseline peaks at 0.906 at $x/c = 0.054$ and likewise has none.

This is the one place in the campaign where a candidate's flow is qualitatively different from the baseline's rather than quantitatively so, and it is what makes C and M the two most expensive candidates at cruise. It is local in span: at $\eta = 0.60$ and 0.40 neither C nor M reverses a single face, and at $\eta = 0.95$ each reverses one face, 0.010 per cent.

A.4 Late cruise, $M = 0.78$, $C_L = 0.544114800210$

Late cruise repeats early cruise station by station. At $\eta = 0.80$ C and M carry the strong shock, 43.6 at $x/c = 0.7325$ and 38.0 at 0.7175 , peak at $-C_p = 1.248$ and 1.228 , and separate behind it: 24.1 per cent of C's upper-surface faces reversed, over $x/c = 0.734$ to 0.994 , and 25.3 per cent of M's, over 0.725 to 0.997 , with both lower surfaces clean. F and H carry the weaker shock, 16.1 and 16.8, and stay attached. W and the baseline have no aft shock. Off $\eta = 0.80$ the picture is the early-cruise one: no reversal on C or M at $\eta = 0.40$ and 0.60 , and one face each at $\eta = 0.95$.

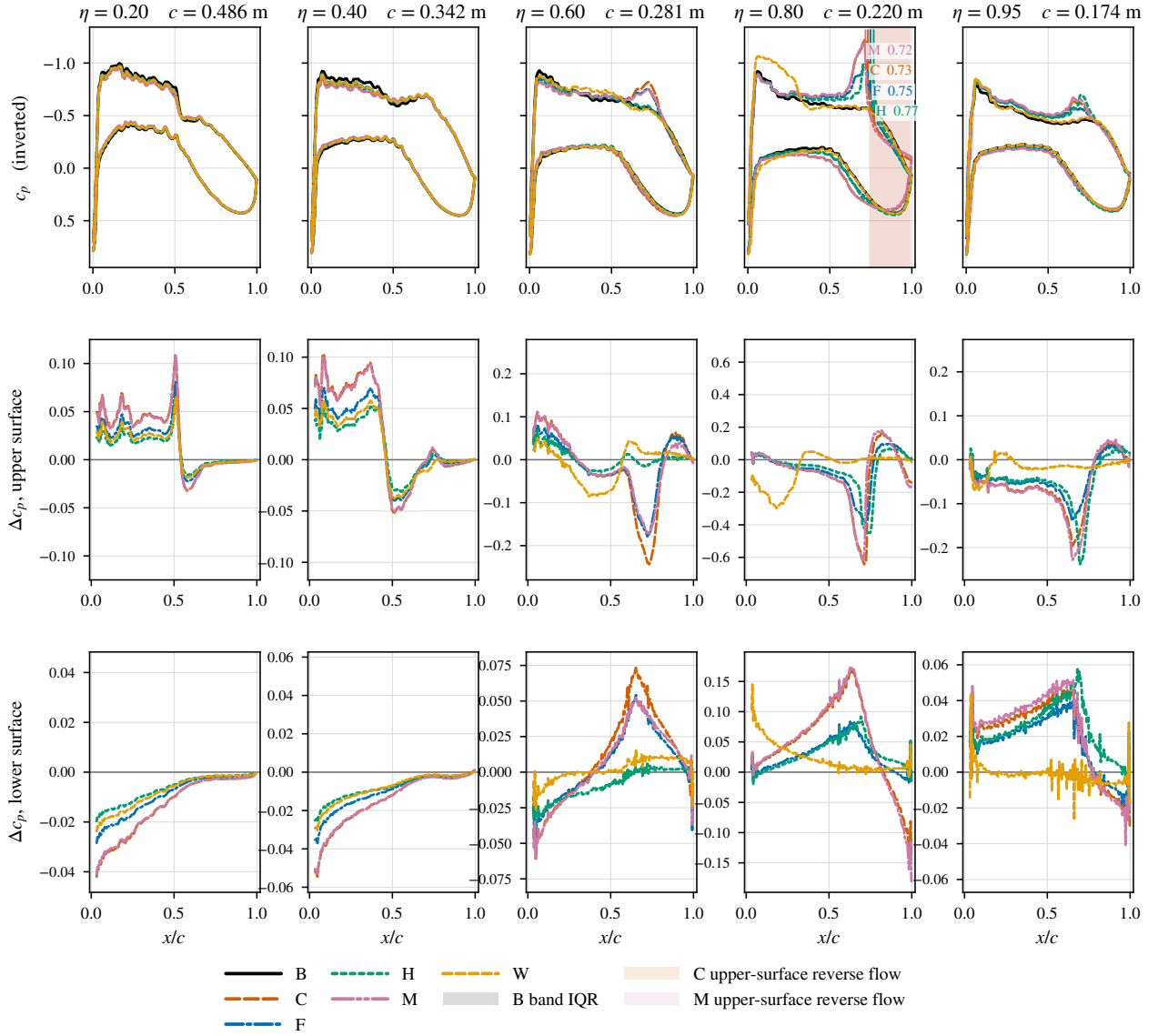


Figure 18: Surface pressure at the same five stations, late cruise, every geometry at its own trim to $C_L = 0.544114800210$; layout as Figure 17.

Provenance. Sections are cut from each case’s own `wingSurface.vtk` at its converged iteration by `scripts/section_slices_from_surface.py`, which also writes the upper/lower flag from the per-face Newell normal, its sign calibrated per case against the more negative mean C_p at mid-chord and cross-checked against median z , refusing rather than guessing if the two criteria disagree. The wind axes are the trimmed leg’s own, read from its `forceCoeffs` history in numeric order of the legs. The cruise surfaces were pulled from the cluster with every file’s checksum verified, and each section manifest records the source case, the surface checksum and the extractor version. The figures are drawn by `fig/make_cp.py` and `fig/make_cf.py`. The station files, their per-file headers (q , p_∞ , local chord, `dragDir`, shear sign) and the extraction manifests are in `data/sections_welded/` for both cruise points, `data/sections_r2/` for B, C, F, H and W at Condition CR and `data/sections_nflag/` for M at Condition CR, resolved per case by `fig/section_source.py`; the superseded set carrying the geometric split is retained beside them in `data/sections/`.

B Surface differences on the wing

Purpose. Section A cuts the wing at five stations. This section gives the same comparison over the whole surface at once: the baseline field, and each candidate minus the baseline, on the upper and lower surfaces together. Twelve panels per quantity per condition, so a feature can be located in span and chord at the same time rather than inferred from five slices.

A difference between two separately meshed wings needs a common support. The wings are meshed independently and share no cells at all: SWB carries 11,334,640 surface triangles against SWC’s 11,356,180, and no face of one corresponds to any face of the other. A cell-to-cell subtraction therefore does not exist. Every face is instead binned onto a common $(\eta, x/c)$ grid of 240×320 cells, area-weighted, with upper

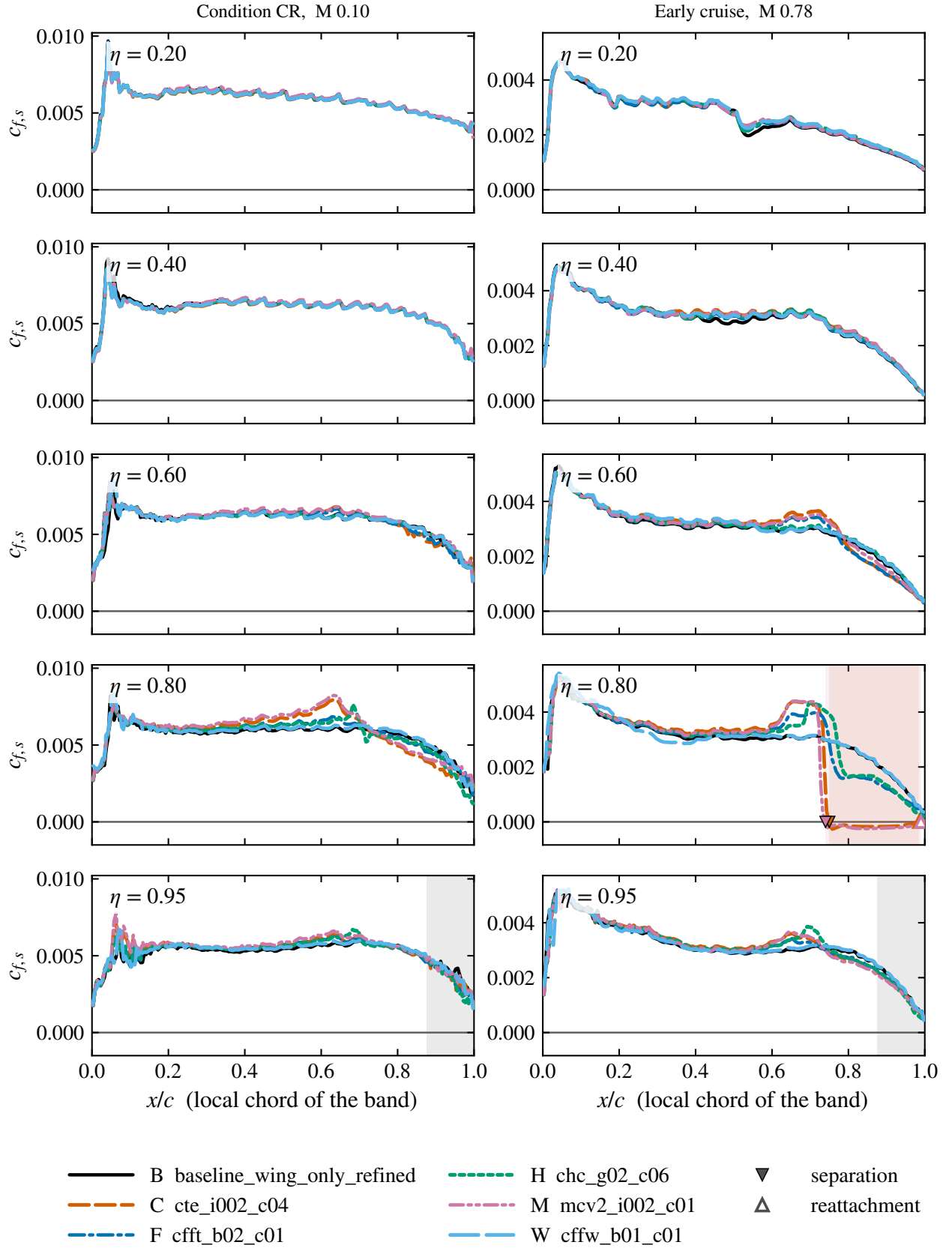


Figure 19: Signed skin friction $c_{f,s}$ at the same five stations, **upper** surface, Condition CR wall resolved on the left and early cruise wall modelled on the right, each geometry at its own trim. Positive is attached and negative reversed, so it is the sign change and not the magnitude that locates separation: the two separated runs at early cruise, C and M at $\eta = 0.80$, are marked with their separation and reattachment stations. Late cruise separates at the same two bands (Section A.4). The grey region at $\eta = 0.95$ is where the aft bin is band-averaged rather than resolved.

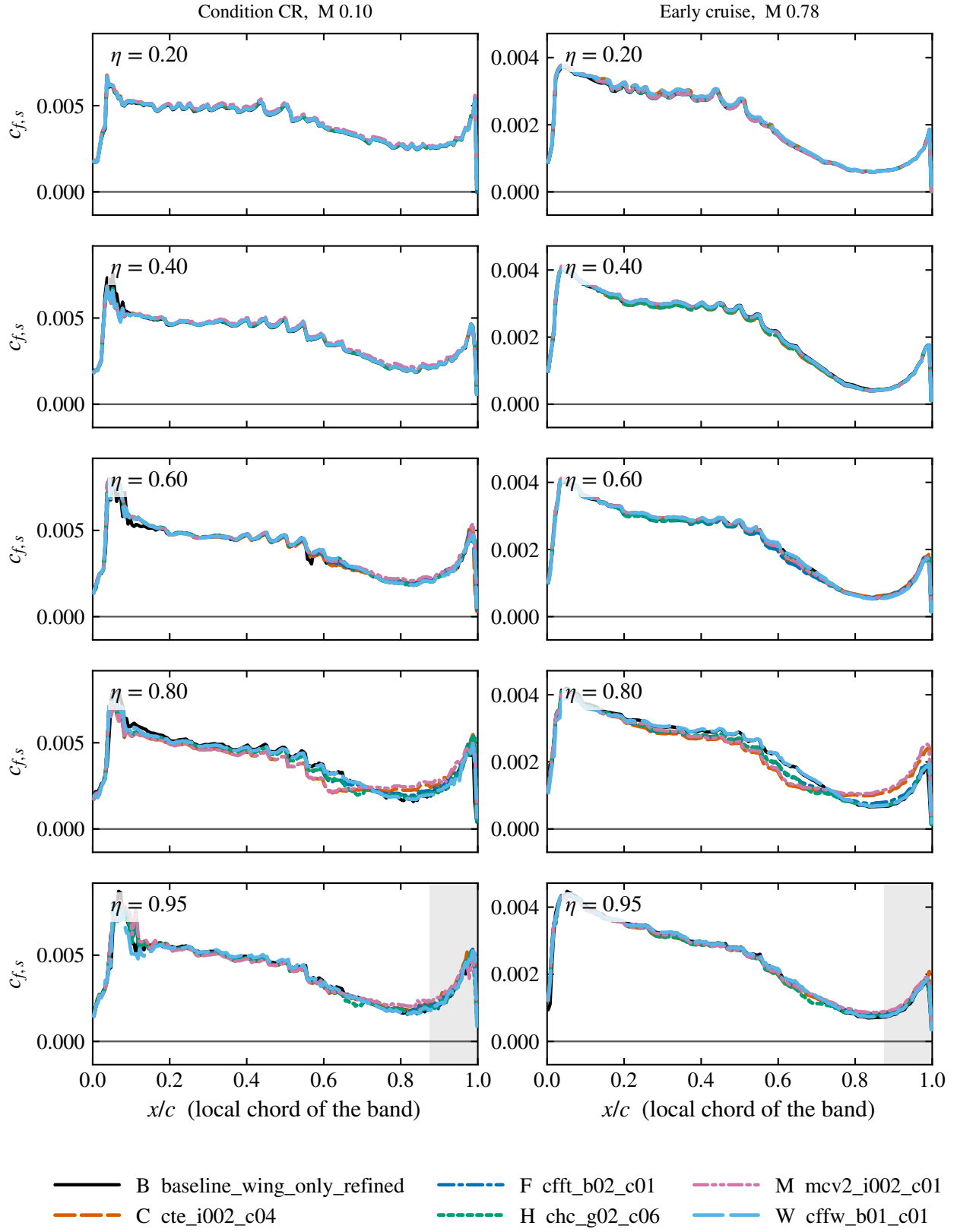


Figure 20: The same, **lower** surface. No lower-surface band carries a separated run at either condition; the traces at the blunt trailing edge are the base region, not boundary-layer separation.

and lower surfaces binned separately using the flag each export carries. That grid is the only support on which a difference between these two wings is defined, and 76,797 of its cells are filled on both surfaces.

The panels are drawn on the planform, not on the grid. An $(\eta, x/c)$ image is a rectangle, and a rectangle silently asserts a rectangular wing to any reader who does not stop to read the axes. Each bin therefore carries its physical position and the panels are drawn on the true swept and tapered planform, matching Figure 2 so the two can be read against one another. Span runs left to right with the root at the left, x increases upward, so the leading edge is at the bottom. The drawn planform reproduces the registered wing to 0.18 per cent in root chord (0.6403 m against 0.639125), 0.2437 against 0.238 in taper, and 1.0036 m of leading-edge sweep. It is a canvas for a field, not a measurement of the planform.

What the figures carry. Column B is the baseline's own absolute field on its own scale. The five candidate columns are candidate minus baseline on ONE shared scale, set at the 99.5th percentile of $|\Delta|$, so their magnitudes are comparable by eye across columns. Blank cells are cells no face landed in, and are never zero. Every panel title carries that case's C_D in counts, its ΔC_D against the baseline of the same condition, and its L/D , read from the same `rans_forces_current.json` harvest as the tables in Sections 5.1 and 5.2, so a figure and a table cannot disagree. Each wing is at its own trimmed incidence at the common target C_L , so every comparison here is at MATCHED LIFT. **A difference forward of the hinge is lineage, not morphing.** The candidates are not identical to the baseline ahead of the commanded region. Measured on the placed surfaces in absolute coordinates, C, F, H and M all depart from the baseline by the same amount, 0.316, 0.184 and 0.079 mm at $\eta = 0.60, 0.80$ and 0.95 , and by exactly zero at $\eta = 0.40$ and inboard; W departs further and differently, 0.366, 1.171 and 0.292 mm. Four candidates sharing one forward shape that the baseline does not is the baseline-lineage difference of Section 2.3 appearing in the meshed surfaces, and it does not cancel in a baseline-to-candidate difference because the two sides sit on different lineages. It is small, 2.6 per cent of the aft displacement at $\eta = 0.80$, but any faint forward feature in the candidate columns should be read as lineage rather than as an effect of the morph.

Null. Differencing the baseline against itself returns exactly zero in every filled cell, for every binned quantity and all three conditions, 76,797 cells each. The renderer also refuses to difference across operating points: it compares condition, Mach, U_∞ , grid size and registered semispan between every pair before subtracting, and stops rather than differencing unlike states.

B.1 Pressure

At Condition CR the candidates are almost free. The pressure differences are confined to the commanded region and are small. The trimmed drag behind each panel is Table 14, and the differences there are small beside the cruise ones but they are not scatter: the systematics this report quantifies, in Section 6.2, total at most 0.024 counts, being window drift of 0.0014 to 0.0028 ct and a trim-residual effect of at most 0.021 ct. Section 7.3 states what a margin of this size does and does not support. At cruise the same geometries separate into three groups. The baseline runs 177.8 counts at L/D 29.76 at early cruise; W saves 0.8, F and H cost 1.8 and 2.2, and C and M cost 12.9 and 14.6. Late cruise gives the same order. The aft-camber candidates carry a recompression shock on the upper surface that the baseline does not, and it shows in these maps as a sharp spanwise band of strongly negative ΔC_p running outboard across the rear of the wing: strongest on C and M, weaker on F and H, and absent on W.

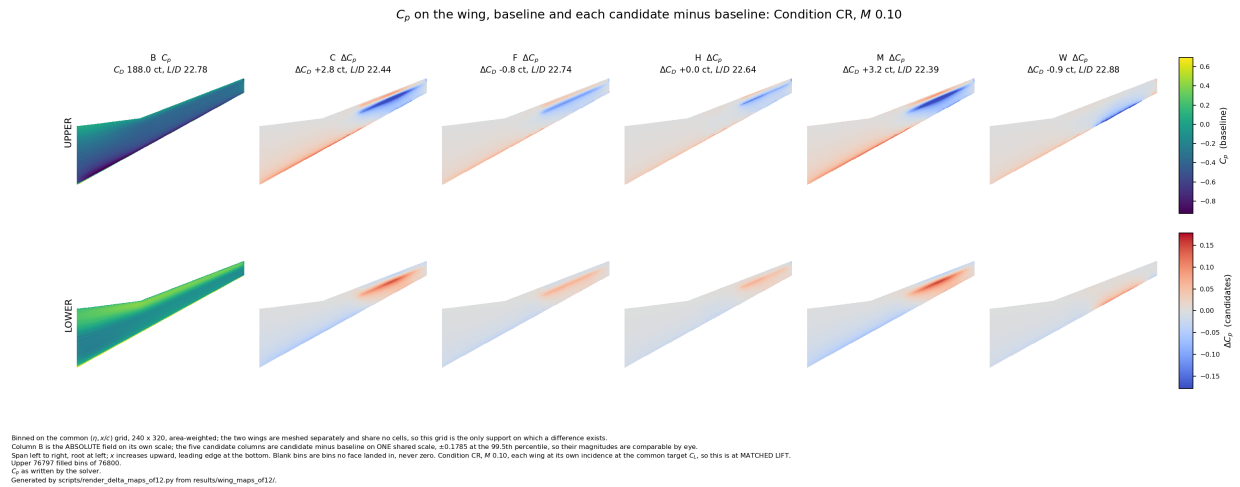


Figure 21: C_p on the wing at Condition CR, $M = 0.10$: the baseline's own field and each candidate minus the baseline, upper and lower surfaces. Each wing at its own trimmed incidence at the common target C_L , so this is at matched lift.

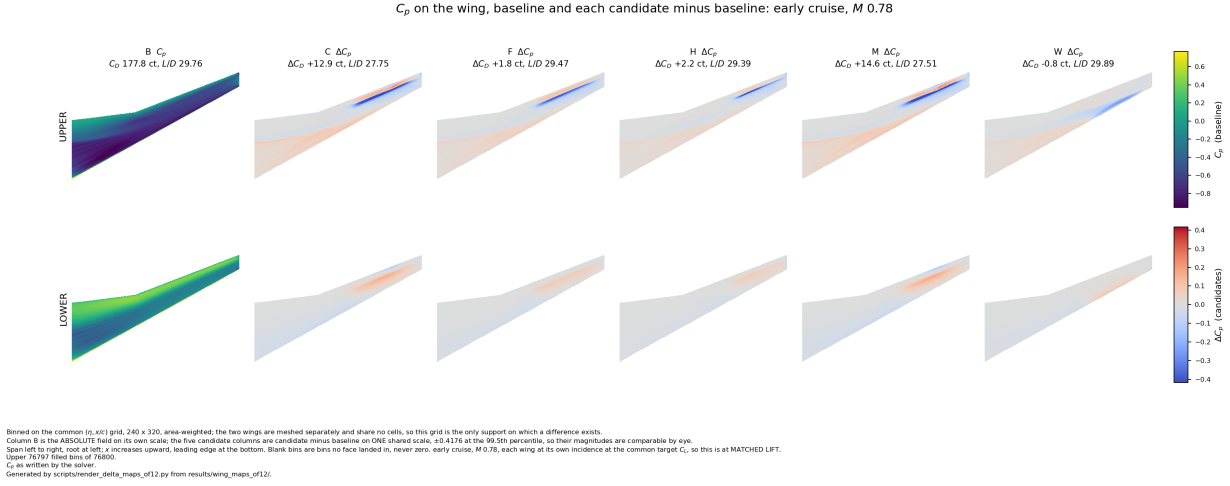


Figure 22: C_p on the wing at early cruise, $M = 0.78$. The band of strongly negative ΔC_p across the rear of C, F, H and M is the recompression shock the baseline does not carry; W has none. The two most expensive candidates on drag, C at +12.9 ct and M at +14.6 ct, are also the two whose boundary layer does not survive that shock; see Section B.2.

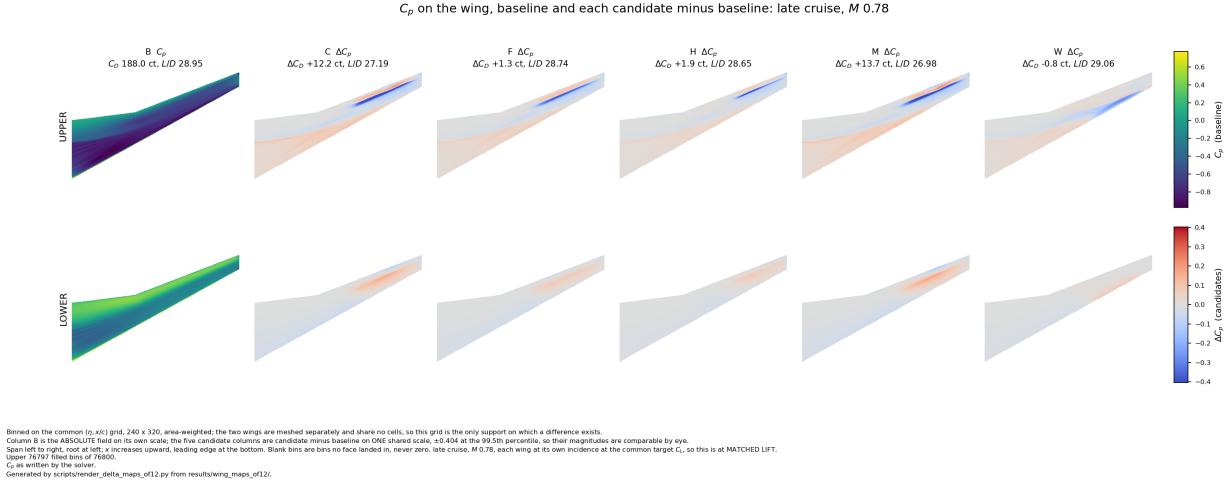


Figure 23: C_p on the wing at late cruise, $M = 0.78$; as Figure 22.

B.2 Skin friction and separation

The signed component, not the magnitude. These maps carry $C_{f,s}$, the component of wall shear along the freestream taken from each case’s own `dragDir`, so negative means reversed flow. The magnitude $|\tau|/q$ is positive by construction and cannot express reversal: a separated region reads as merely small and is indistinguishable from a thinned but firmly attached boundary layer. That distinction is the whole question here, because F carries the same shock as C and M with a visibly weakened boundary layer behind it and does not reverse.

The sign is calibrated, never assumed. Taken raw, the first case reported 99.18 per cent of its faces reversed, which is not a separated wing but an inverted convention: OpenFOAM’s `wallShearStress` opposes the freestream where the boundary layer is healthy. Rather than hard-code a minus sign, the sign is measured per case against faces whose attachment is known independently of this field, namely inboard mid-chord upper faces at $\eta = 0.15$ to 0.35 and $x/c = 0.35$ to 0.65 . It came out one-signed on all eighteen trims, with about 260,000 calibration faces each.

Nothing separates at Condition CR. All six geometries reverse at most 0.013 per cent of their upper-surface bins, which is 0 to 10 bins of 76,797 at the root junction and the blunt trailing edge, and is present on the rigid baseline too. The six are indistinguishable from one another and from the noise floor.

At both cruise points two candidates separate and four do not. Measured over the whole upper surface at early cruise: C reverses 1.520 per cent of its bins and M 2.122, against 0.004 for the baseline, F and H and 0.000 for W. At late cruise C reverses 1.803 and M 2.335, the other four the same as at early cruise. That is a factor of some 400 between the two groups, on the same grid and the same code path. The lower surfaces reverse 0.000 per cent on every geometry.

The maps agree with the section cuts, on a different support. At $\eta = 0.80$ at early cruise the binned maps give 23.28 per cent of bins reversed for M and 22.97 for C, against the face-level section extraction’s 23.86 and 23.27 per cent at the same station; at late cruise 25.16 and 23.91 against 25.28 and 24.06. Both methods return exactly zero for B, F, H and W at both cruise points. Two independent code paths, one binning onto a common 240×320 grid and the other counting raw faces in a band, agree to within 0.6 percentage points and

agree exactly on which cases separate. The whole-wing figures above are smaller only because the whole wing is a much larger denominator: the reversal is real and confined in span, not damped by the binning.

Reduced skin friction is not separation, and the contour is what separates the two. A difference panel shows that the shear FELL, not that it REVERSED. The black contour is each case's own $C_{f,s} = 0$, so flow inside it is reversed and the contour's ABSENCE is itself a result. It is drawn only where at least 0.1 per cent of a surface's filled bins are negative, because every case carries a handful of negative bins at the root junction and the trailing edge; that threshold sits about eight times above the worst noise (10 bins) and fifteen times below the smallest real signal (1167 bins, for C at early cruise). At Condition CR no panel draws a contour. At both cruise points only C and M do.

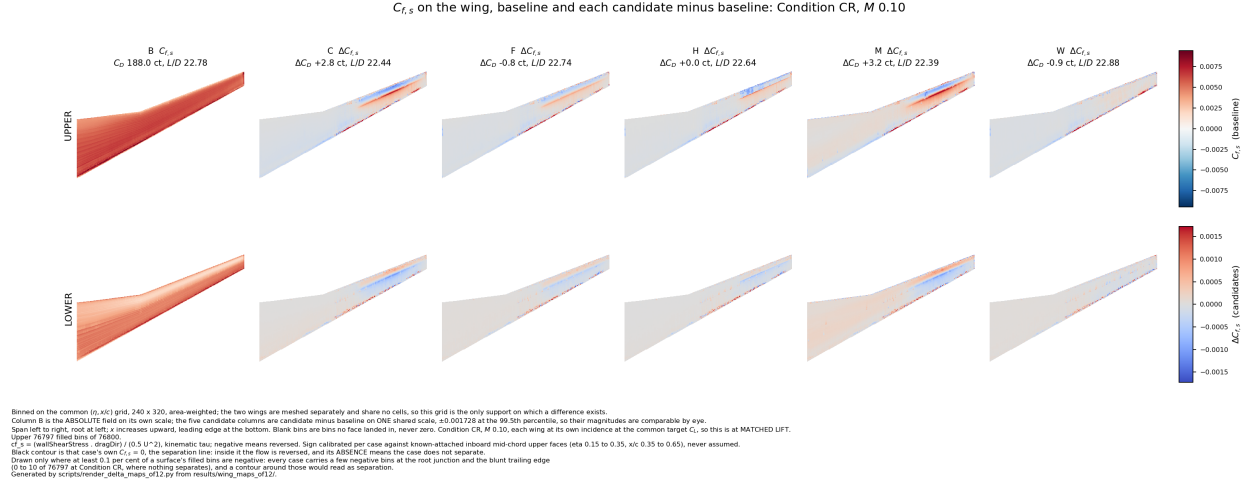


Figure 24: $C_{f,s}$ at Condition CR, $M = 0.10$. No separation contour appears on any panel: every geometry is attached everywhere, and the candidates differ from the baseline only in the magnitude of the shear over the morphed region.

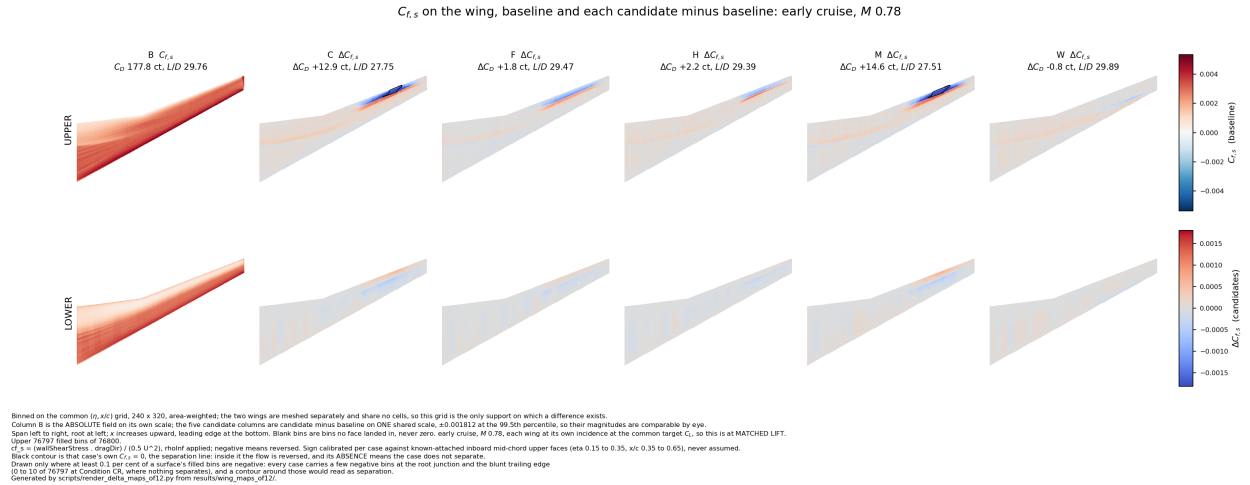


Figure 25: $C_{f,s}$ at early cruise, $M = 0.78$. C and M carry a closed separation contour over the rear outboard wing; F, H and W do not, although F and H show a fall in shear behind their weaker shock. The two most expensive candidates on drag, C at +12.9 ct and M at +14.6 ct, are the two whose boundary layer does not survive it.

References

- [1] Morgan, H. L., and Paulson, J. W., *Low-Speed Aerodynamic Performance of a High-Aspect-Ratio Supercritical-Wing Transport Model Equipped With Full-Span Slat and Part-Span Double-Slotted Flaps*, NASA TP-1580 (L-13201), December 1979. NTRS 19820007142.
- [2] Morgan, H. L., *Model Geometry Description and Pressure Distribution Data From Tests of EET High-Lift Research Model*, NASA TM-80048, February 1979.
- [3] *Experimental Data Base for Computer Program Assessment*, AGARD AR-138, 1979. Onera M6 wing, run 308; the validation case for the cruise numerics used here.
- [4] Zheng, L., *ARGUS Aerodynamic Optimization of a Morphing Outer Wing*, internal ARGUS design-space-optimisation report, July 2026, with the far-field C_{Diw} audit delivery of 19 August 2026. Table 18 is the cruise operating-point table referred to in the caption of Table 2.
- [5] *OpenFOAM*, version 12, OpenFOAM Foundation. All cases run `foamRun` with the `incompressibleFluid` and `fluid` modules.

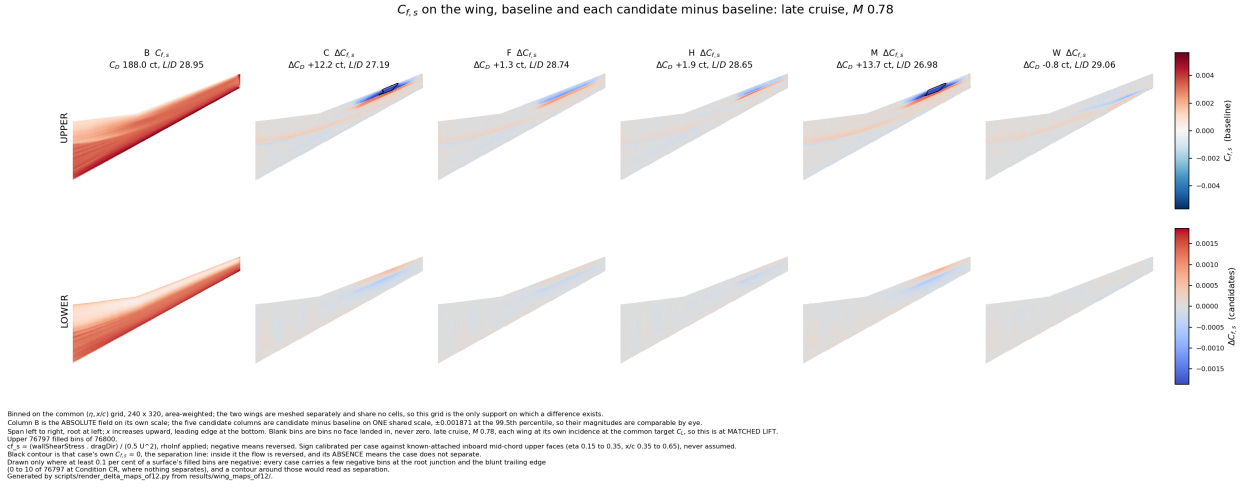


Figure 26: $C_{f,s}$ at late cruise, $M = 0.78$; as Figure 25.